MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLAGE OF INFORMATION AND COMMUNICATION TECHNOLOGY

DEPARTMENT OF INFORMATION SYSTEM AND TECHNOLOGY

DIPLOMA IN INFORMATION AND COMMUNICATION TECHNOLOGY

NTA LEVEL 6

FINAL PROJECT REPORT

PROJECT TITLE : DEVELOPMENT OF ONLINE SYSTEM FOR SEARCHING AND APPOINTMENT OF ADVANCES

PROJECT TYPE : WEB BASED

NAME OF SUPERVISOR : Mr. AZIZI SINGA

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JULY, 2023

CERTIFICATION

The undersigned certify that has read, and hereby recommend for acceptance by Mbeya University of Science and Technology project tittle “ONLINE SYSTEM FOR SEARCHING AND APPOINTMENT OF ADVOCATES” In partial fulfillment of the Ordinary diploma in Information and Communication Technology at Mbeya University of Science and Technology (MUST) under the department of information systems and technology in the college of information and communication technology, As submitted by DEVOTHA A. MABULA and CHEREHANI K. MAKIGO.

…….

Supervisor’s Name

……

Signature

……..

Date

DECLARATION

We declare to the best of our knowledge that, this project of "ONLINE SYSTEM FOR SEARCHING AND APPOINTMENT OF ADVOCATES" as presented here, as a partial fulfillment to the award of ordinary diploma in Information and Communication Technology is our original work and it has neither been copied from anywhere nor presented elsewhere.

…….. ……

ADEVOTHA A. MABULA CHEREHANI K. MAKIGO

……

Date

DEDICATION

We dedicate this work to our beloved nation of Tanzania and our precious university as Mbeya University of science and technology (MUST). The work and the effort are all dedicated for the best of our nation and citizens.

Also, we dedicate this work to our head of department and all facilitators concerned in this work. The people who decided to grind with us for the completion of this task and all their times they gave us till the completion of this project.

Finally, we dedicate this project work to our lovely families who have never got tired for inspiring us to make it well, and without forgetting our fellow students show gave us their support and advice till the completion of this project work.

ACKNOWLEDGMENT

We raise our hands hardly, open our heart widely, and say thanks to the Creator who gave us power and energy to carry out this project work without becoming ill or injured. Many thanks and gratitude to God who gave us each and everything that has made us be where we are today.

Sincerely we also say thanks to the United Republic of Tanzania, the President, ministers and all ministries that have given us the peace and harmony for us to work on our land without fearing anything. We also say thanks for all the services they gave us to accompany us in this practical training time. The roads were really good and favorable for us to make this project.

We also acknowledge our precious university of Mbeya university of science and technology (MUST) for all the disturbance we made to it but still they endured till the completion of this project. We cannot pay them but we hope God will make it the best for them. We are glad they aided us in many ways.

Thanks to our department and all the facilitators for their untied support and precious time they gave us till the completing of this project. We can't count the problems we brought in their families for the late coming due to projects and all the night calls we called them for misunderstood concepts.

Thanks to our families for their generous hearts and understanding minds for whole project time they gave us till completion. We are happy to have these families in this world and we always pray for all families who have assisted us in any way until the completion of this project.

Without forgetting our fellow students all around the university and our fellow classmates for their tremendous presence and support in completing this project. May God bless you and give more in your journey to keep supporting others on their way here. We are glad you crossed our paths and special thanks to you all.

Finally, we drop our thanks to our own hearts, mind and body for all the times being there when needed. We say it was tough but our hearts never gave up, our minds never stopped thinking and our bodies never get tired and kept sharing until we have completed this project. We have learned a lot in this project.

ABSTRACT

Online System for Searching and Appointment of advocates are used widely in any situation where any one need to contact with Advocate and the advocate also manages their daily schedule and client's query. This project's perspective, we would have given a name and that's "Online System for Searching and Appointment of Advocates". Using this website client can view the advocate profile, search advocate, view advocate schedule, send appointment request, send message to advocate and advocate can reply client message.

LIST OF ABBREVIATIONS

AJAX - Asynchronous JavaScript and XML

CoICT - College of Information and Communication Technology

HTML - Hypertext Markup Language

ICT - Information Communication Technology

IT - Information Technology

MUST - Mbeya University of Science and Technology

MySQL - My Structured Query Language

NTA - National Technical Award

OSSAA - Online System for Searching and Appointment of Advocates

SDLC - System Development Lifecycle

UI - User Interface

UML - Unified Modeling Language

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CHAPTER ONE
INTRODUCTION

1.1 Introduction

This project is designed to convert the advocate paper system into online system that not only facilitates advocates to get soft catted result of case registration, hearing time tables and save and update information about clients and their cases as well as facilitate clients to check their case related information online. It will also help to get rid of paper form filling.

Advocates are often busy with paperwork and client meeting, making schedules invaluable tools. Look into the importance of scheduling for advocates and explore methods such as to do lists, digital, reverse scheduling and other electronic tools.

1.2 Problem statement

When a client his/ her live in Mbeya region he/she may go to advocates office for searching and booking appointment until an Advocate can reapply the client applications offline. When a client completes his/her applications in the advocates office he/she will need a confirmation that will prove that he/she has carried out the booking appointment.

The problem with the current systems is that do not tell us where to find the relevant advocates according to our cases that we have. They only want common people to search for the advocates for whom they know by their names. The system is difficult to use, needs the client to go physically directly to the Advocate's office in order to get law services.

A very effective way to reduce long service and queue wait time and would be to automate the process of searching, booking and scheduling an advocate they want. This way only clients that have been scheduled to see the advocate would be available at a time and it'll help boost the service experience of the clients and the work morale advocates as the clients can now make better use of their time.

1.3 Proposed system

We have proposed this system in order to solve the above listed problems we have suggested we should get an online system that will serve all the process and solve the problem at a solely level. The proposed system has a target of reducing time wastage and reducing cost in this current searching and booking appointments issuing system.

Online system for searching and appointment issuing is our best solution for these problems. If you observe clearly, you will have a need to issue the online system for searching and appointment on the web system in order to reduce queue along the offices and increase efficiency in the online searching and appointment of advocate issuing system. We think this is the best solution we will ever need in order to solve all these problems.

1.3.1 Advantages of the proposed system.

- There will be no queue at all since the clients will request the searching and appointments on the website.
- No need to use money for travelling from home to the advocates offices for the sake of searching and booking appointments unless otherwise.
- Easing the process of job application since the clients will use the soft copy sent to them.
- Increasing the efficiency among workers by allowing less tiresome works like carrying files of clients.
- Less time since the response is done in the website to avoid.
- Serving many clients at a time by confirming their request online.

1.3.2 Disadvantages of the proposed system.

- Less workers will be needed for the task to be complete.
- Decreasing people flow to our advocate's office for the sake of booking appointments.

1.4 Objectives

Objective is a set of information based on facts rather than feeling or opinions. The project we are carrying out has some objectives or a realistic goal on how it will come into effect as soon as possible. In this project we have two parts of the objectives that are:

1.4.1 Main Objective

The main objective this project is **to Develop Online System for Searching and Appointment of Advocates**.

1.4.2 Specific Objectives

1. To design the website for online searching and appointment of advocates.
2. To design a database for data storage.
3. To design a prototype of the proposed system.
4. To implement the system.

1.5 Scope of the Project

The scope of this project shall end on implementation of the system.

1.6 Limitation of the study

- Absence of internet connection does not provide access to the website for the clients to request appointment from our system.
- The customer's costs during the advocate services are not constant.

1.7 Significance of the project

- Will create the open door for employed and unemployed advocates.
- The system will be available twenty-four hours/seven days a week allowing for scheduling at a time that best suits the customer
- Customers will be able to view the Advocate's availability for appointments and schedule accordingly.
- Advocates will be able to see the appointment from the customers and then they will be able to cancel or approving them and providing feedback to the customers.
- Customers will be able to request for the appointment for the advocates.

1.8 Case study of our Project

The study will focus on developing an Online System for Searching and Appointment of Advocates to allow different people to communicate and interact online with Advocates. The study will be conducted in Mbeya Region.

CHAPTER TWO
LITERATURE REVIEW

2.1 Introduction

The literature review acknowledges the work of previous researchers, and in so doing, assures the reader that your work has been well conceived. It is assumed that by mentioning a previous work in the field of study, that the author has read, evaluated, and assimilated that work into the work at hand.

The current systems do not tell us where to find the relevant advocates according to our cases that we have and required the clients to go physically direct to advocate.

2.2 Review of related works

This chapter consists of conceptual reviews and thematic reviews of related literature that help to understand the factors affecting existing advocate information systems and conditions for the successful implementation of an online advocates searching and appointment system.

2.2.1 DuLaw

Founded by (Dudumizi, 2010), is an established web design and Mobile Development Company at Dar Es Salaam, Tanzania. DuLaw is an all in one, affordable and intuitive Law practices management system which allow Law firms to manage day to day activities anytime anywhere. Features including; compatible with Tanzania legal frameworks, customer centric and total collaborative tool. DuLaw is an all in one, affordable and intuitive Law practices management system which allow Law firms to manage day to day activities anytime anywhere. Features including; compatible with Tanzania legal frameworks, customer centric and total collaborative tool. Our system will allow the Advocate to register in the system itself and be able to enter his important information that will help his client get to know him better.

2.2.2 Sheria Kiganjani

Chris Bwemo is the founder and network champion for **Sheria Kiganjani.** Sheria Kiganjani is an online legal digital platform which enables an individual, (Bwemo, 2010), a student layer or a group of people to access legal information as well as materials despite their location. Our system will allow client to enter and search for the Advocate they want based on the system itself.



Figure 2.0.1 sheria kiganjani logo

2.2.3 The Advocates Global

Founded in 2017 By Mario P. Proenca, CEO and Managing Partner. The advocates Global is the leading premier directory of Law Firms and advocates global Elite, (Mario P.Proenca, 2017). Our system will allow the advocate to register on the system itself and his/ her to create a profile.

2.2.4 Advocate Wangu

Advocate Wangu is Kenya's trusted "Do it yourself" online platform for creating legal documents. The process is simple, quick and affordable such as choose an issue, provide a short brief of the problem or not, make a payment and speak to an advocate for services, (Anon., 2012) . Our system will allow the client to request a meeting, booking appointment with their Advocate and see the relevant Advocate calendar of events.

2.2.5 Law Connect

Jared D. Correia, Esq. is the CEO of red Cave Law Firm Consulting, which offers subscription-based law film business management. The primary focus here is on client discovery and booking processes and integrates with a wide range of third-party systems. Features; No free option, cost 20$ per month, clients can book appointments via your website (Jared D. Correia, n.d.). Our system will be cheap which will help everyone to be able to use the system more easily.



Figure 2.0.2 Law Connect to find your Advocate in US

2.2.6 In the present system the problems are: -

- ❖ Manage advocate schedule time
- ❖ In order to find the advocate, you need to know them first
- ❖ To find best advocate according to problem
- ❖ This requires a lot of time and effort

2.3 Advocate Appointment Booking

Advocate provides services to people around the world, advocates usually experience an influx of clients daily. Now day's advocates are looking for systems like booking systems to automate appointment booking that will connect them with their clients so that they can serve them more Easily. These systems help to curtail prevalent issues of long wait and service times often experienced by clients. They also help to improve customer satisfaction, patronage, and control the wastage of time.

2.4 Benefits of Online Appointment Booking Systems

There are a lot of advantages that come along with implementing appointment booking systems to automate appointment booking between advocates and their customers. Some benefits of law information as indicated by (Reddy, 2020), are discussed in the following part of this section.

a. Boosts client's satisfaction – In the conventional appointment booking technique, advocates or their customers must manually schedule an appointment and send reminders indefinitely. When an advocate office adopts an appointment scheduling program, advocate can pay more attention to their customers' requirements. Advocate can focus on providing the client with the care they need upon arrival at the advocate's office, giving them time to do something other than making an appointment by hand.
b. Better ability to process and store client data – Before the introduction of law information systems, client's information was recorded in a book or manually entered into a system. All clients' information can now be maintained, updated, and managed correctly through the use of an online appointment scheduling system.
c. Helps to save time – An advocate's visit can be made easier by arranging it online. It eliminates the need for clients to remain in a waiting room for extended periods. Clients save time with less messy and more scheduled visits to the advocate. As a result, client's satisfaction is increased and long-term loyalty is guaranteed.

CHAPTER THREE
METHODOLOGY

3.1 Introduction

The approach that are going to use is OBJECT ORIENTED METHODOLOGY. We choose this method because is the way to manage a project by breaking it up into several phases. It involves constant collaboration with stakeholders and continues improvement at every stage. One the work begins, teams' cycle through a process of planning, executing and evaluating. Continues collaboration is vital, both with team members and project stakeholders. This methodology which an entirely different approach to information systems includes a class diagram, UML sequence diagram, use case diagram, entity relationship diagram and context diagram.

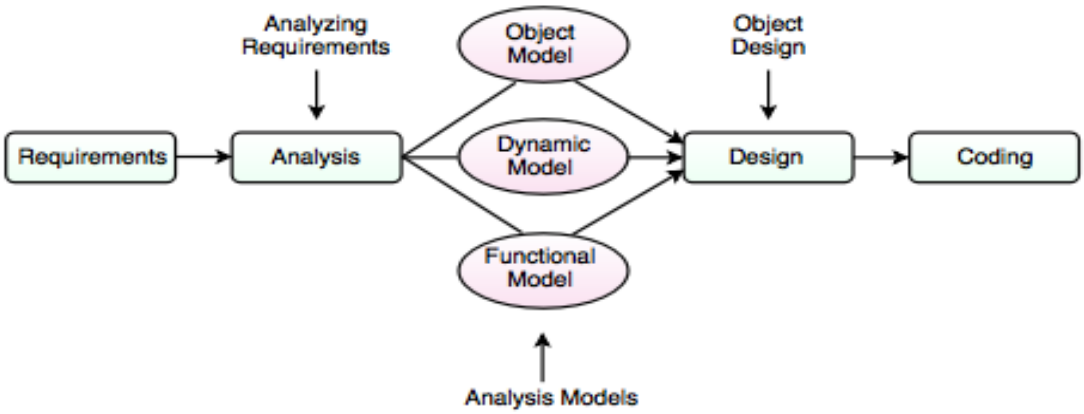


Figure 3.0.1 Object Oriented Methodologies

3.2 DATA COLLECTION AND DATA PRESENTATION

3.2.1 Introduction

This is referring to the general ways into which the information gets collected for the conducting of the project. There are several ways into which data and information can get collected.

We are going to use interviews, questionnaires and document reviewing in the data collection process for our project.

REQUIREMENTS	METHODS	TOOLS
Data collection	Questionnaire	Paper Questionnaire
	Interviews	Face to Face Discussion
	Document Reviewing	Documents
Population sampling	Simple random sampling	Pick random

Table 1.0.1 Data collection requirements, methods and tools

3.2.2 Interviews

Thus, it is very suitable for this study to get rich and detailed information about market structure, efficiency, regulations etc. from practitioners' viewpoints.

Interviews creates a room for important questions meaning that some official will have no need of questionnaires because it will be limiting some answers. The interview is helpful in the variable setting process and answering sub- questions in the study. Besides, the information from interview is used to justify some of the relationships in the quantitative result.

3.2.2.1 Importance of Interviews

a) They allow more detailed questions to be asked
b) They achieve a high response rate
c) They are useful to obtain detailed information about personal feelings of the farmers.
d) Enable us to collect non-verbal data

3.2.3 Questionnaires

Questionnaire will give us more chances to reach many clients at a short period of time.

3.2.3.1 Importance of Questionnaires

a) Less time consuming
b) Generates a large amount of data
c) Format is easily quantified and analyzed
d) Gives respondents enough time to answer the questions.

3.2.4 Document Reviewing

Document reviewing will give us the important keynotes in the development of our complete system

3.2.4.1 Importance of document reviewing

a) To ensure accuracy in your project data
b) To check for completeness of documentation
c) To ensure that the documentation is compliant with standards
d) To verify that the documentation meets the requirements of your project

3.2.5 DATA PRESENTATION

3.2.5.1 Introduction

This chapter provides an overview of the implementation of the system design specified in the design document. A detailed explanation of the database implementation, user interfaces, and application logic would be discussed in the following parts of this chapter.

3.2.5.2 Database Implementation

XAMP Control Panel was used to implement the backend for the System. The data generated from the system were stored as Database in unique documents grouped as a collection.

3.2.5.3 Authentication and authorization

User registration with the system is handled by XAMP Control Panel authentication services. The users register with a verified email and a password of their choice that is at least 6 characters long. After every successful registration, a user object with display Name, email, password, AUTH Token, and user Role properties is created. The user Role property is used to control access to data.

3.2.5.4 Database

Data generated in the app is stored in XAMP Control Panel Database (MySQL). The database's architecture requires you to store data or data sets as documents, and a group of identical documents is stored as a collection of documents. Four different collections of data were stored in our database.

i. User data – User information collected when new users are registered with the system is stored in the database. This information includes the user name – string, matric number – integers, and email – string, photo string and the role (client, advocate or admin) - string. A unique id is also created to uniquely identify each user. Each user's data is stored in a document with its id identical to the user's id and all the user documents are stored in a user collection. The figure below shows how the user object is stored in the database.

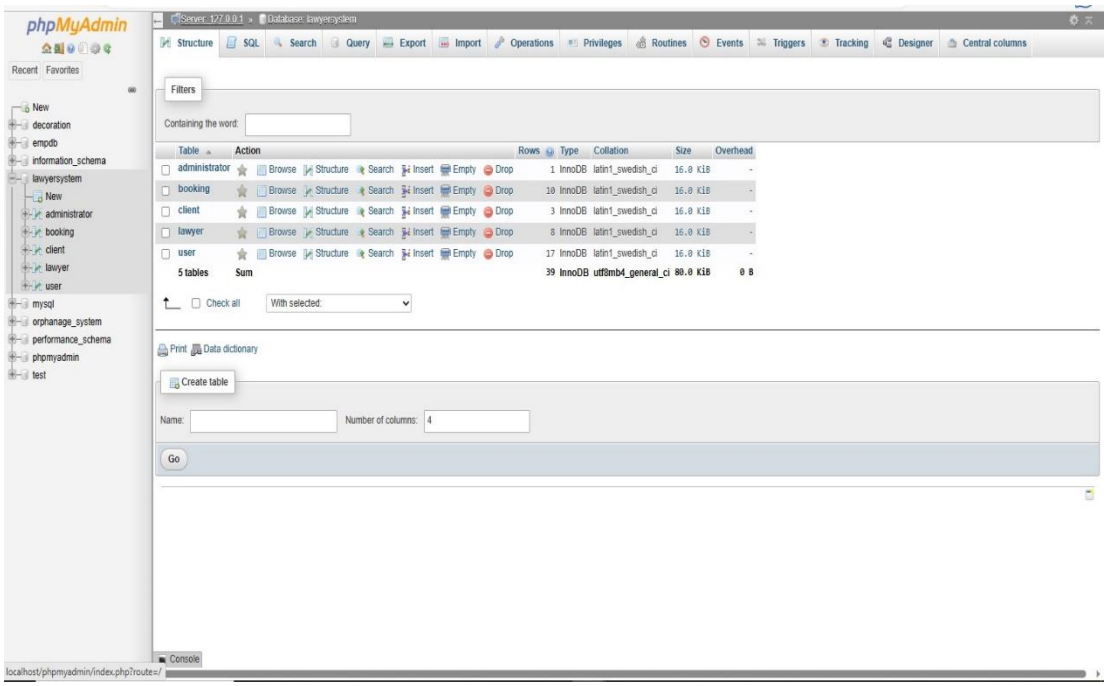


Figure 3.2 Database of the system

ii. Appointments booked by the clients in the system. Each document is an array of all the appointments that have been booked by a user and it's uniquely identified by the users' id. Each appointment has properties of appointment Time – string, appointment Date – string, appointment Status – Boolean, and appointment – Date used to uniquely identify each appointment. Appointment log is illustrated in the figure 3.3

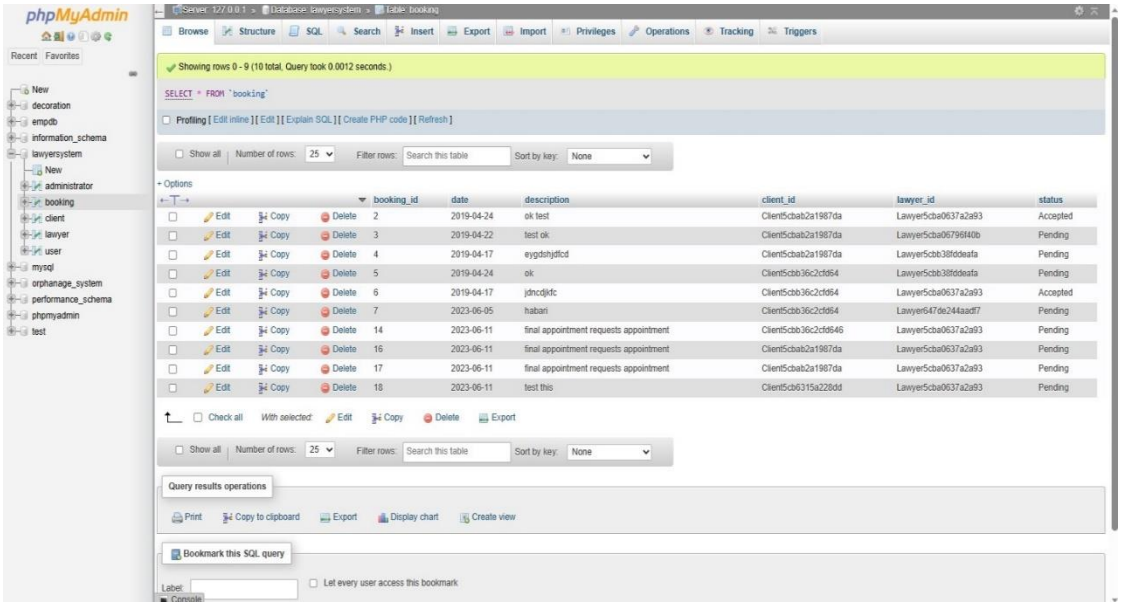


Figure 3.3 appointments booking by clients in the system

3.2.5.5 Appointment booking

The appointment booking page can be used by clients to book advocate appointments. The appointment booking screen responds to on-tap events by the client and prompts them with modals that help them select their preferred date and application times. An alert box with a success or failed message pops up and informs the user about the appointments current state depending on whether the appointment was successful or not.

3.2.5.6 System analysis

Data analysis for quantitative and qualitative studies involves critical analysis and interpretation of figures and numbers, and attempts to find rationale behind the emergence of main findings. A set of analytical system used to assist with analysis of quantitative data. The tools used for data collection and data analysis are interview and MS Office Excel. A sample some of 10 people were taken for the case study to answer the interview. This sample size was used on the set of questions that justified the need of the online system for searching and appointment of advocates to be implemented.

3.2.5.7 Data presentation

In the process of data visualization, graphical methods were selected to represent the data. The histogram graph method has been used to present the analysis of the data for various questions. With reference to the data and corresponding numeric values used in the statistical analysis can be found on the sample interview results. The analysis and interpretation of data is carried out in three phases.

i The first part, which is based on the results of the observation, deals with a quantitative analysis of data.

ii The second, which is based on the results of the interview and Questionnaires is a qualitative interpretation.

Phase One: Analysis of observation

The observation was done in different online system that concerns with advocate s system including “mjue wakili” application and “sharia Kisangani”, all of these applications they don’t have a feature for appointment and scheduling a time table for an advocate, so it implies that there is a need of developing a proposed system.

Phase two: Qualitative Interpretation of results

Analysis of the Interview was to fill the gaps left in the observation and to get the insights of existing rental services, and to get requirements from the users. The qualitative approach was used. An individual structured interview with respondents from local areas was conducted, essential information was obtained. The information collected was presented in a narrative form that includes the description and analysis of data. From the interview, several questions had found answers. Answers that are very vital to the development of the proposed system.

User Interview Questions

1. Please tell me about yourself and your relationship to Law firm.

2. What are the most important tasks you or other people need to perform in using project website or application?

3. Please tell me about your relation to the online system for searching and appointment of Advocates?

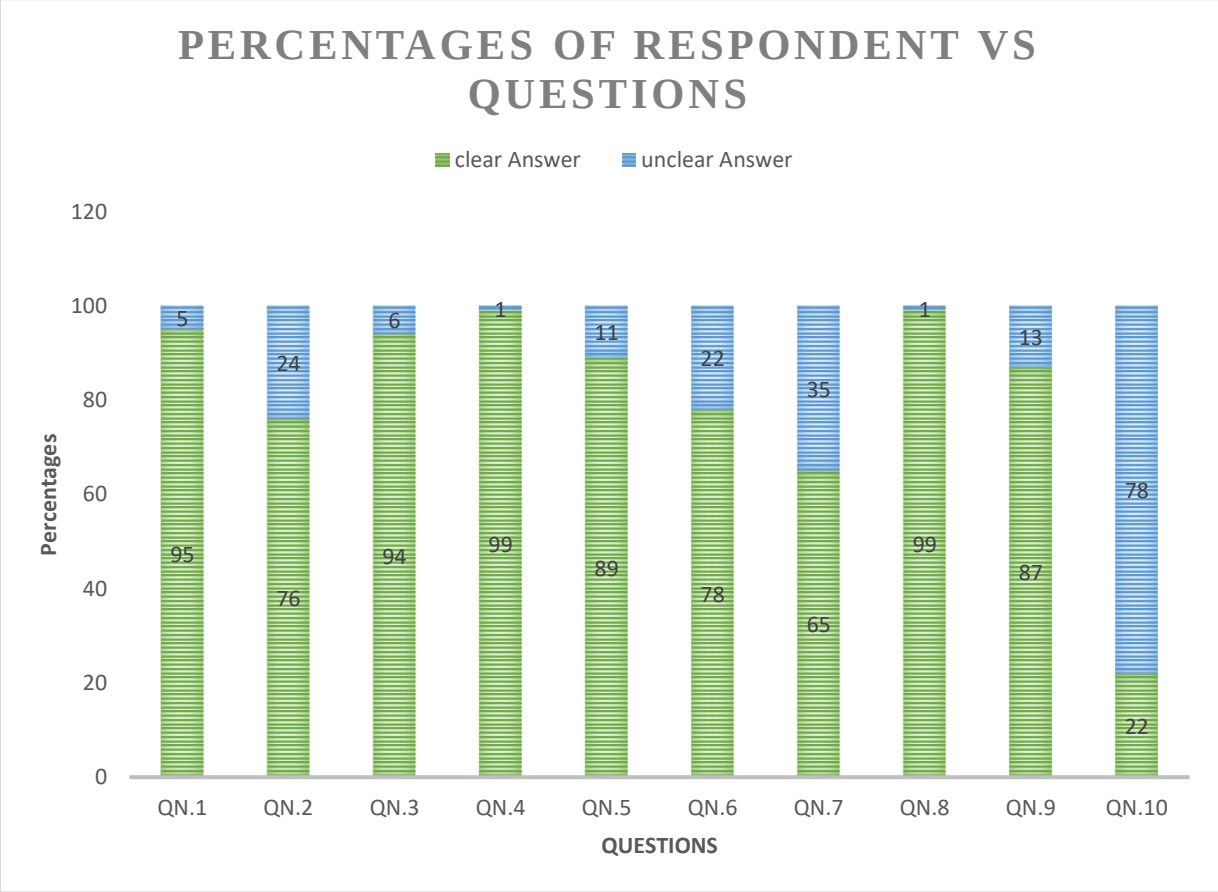
4. What are the most important tasks you or other people need to perform in using website or application?

5. What devices do you typically use when visiting a website or application?
6. Do you or did you in the past use other websites and resources for the same purpose as a website or application?
7. What do you like about the current website or application?
8. What don't you like about the current website or application?
9. What improvements could be made to make project goal easier or better?
10. What haven't we asked you today that you think would be valuable for us to know?

3.2.5.7.2.2 PRESENTATION OF ANALYSIS OF DATA IN THE INTERVIEW

QUESTION NO:	TOTAL PERCENTAGE	PERCENTAGE OF CLEAR ANSWERED	PERCENTAGE OF UNCLEAR ANSWERED
1	100	95	5
2	100	76	24
3	100	94	6
4	100	99	1
5	100	89	11
6	100	78	22
7	100	65	35
8	100	99	1
9	100	87	13
10	100	22	78

Table 3.0.1 Table of the result from the interview



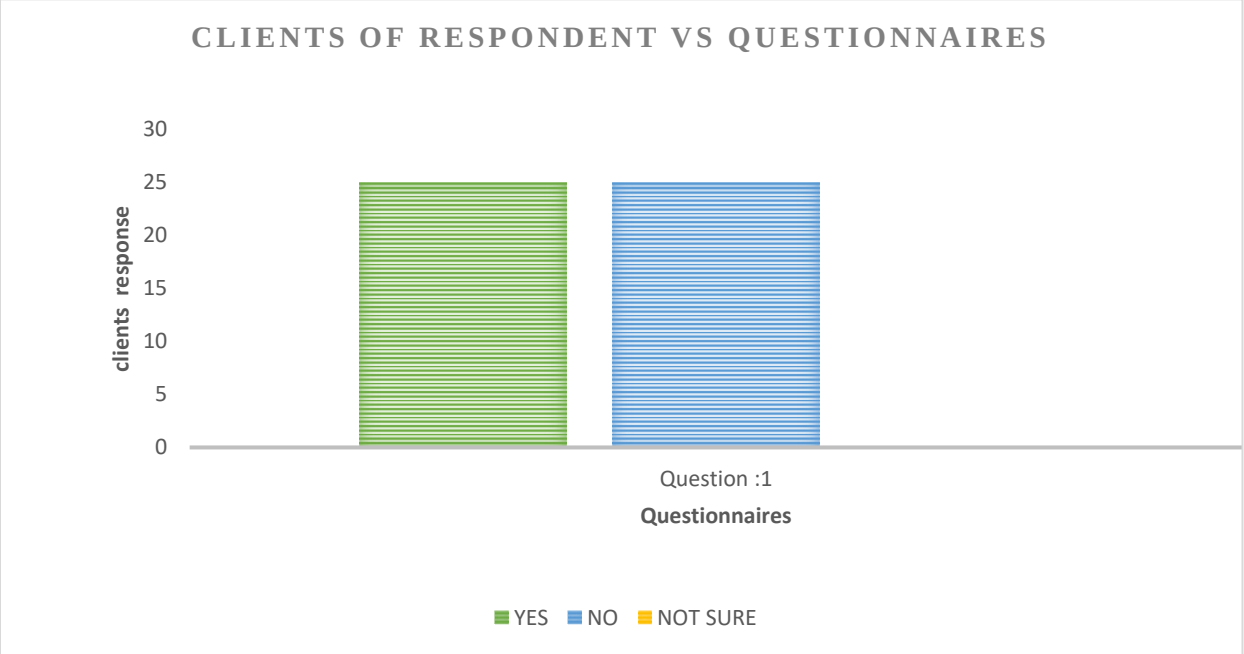
Questionnaire Sample Questions.

QUESTION NO:	TOTAL CLIENTS	CLIENTS ANSWERED (YES)	CLIENTS ANSWERED (NO)	CLIENTS ANSWERED(NOT SURE)
1	50	25	25	0
2	50	35	10	5
3	50	45	3	2
4	50	3	42	5
5	50	41	4	5
6	50	44	3	3

Table 3.0.2 Table of the result from the Questionnaire

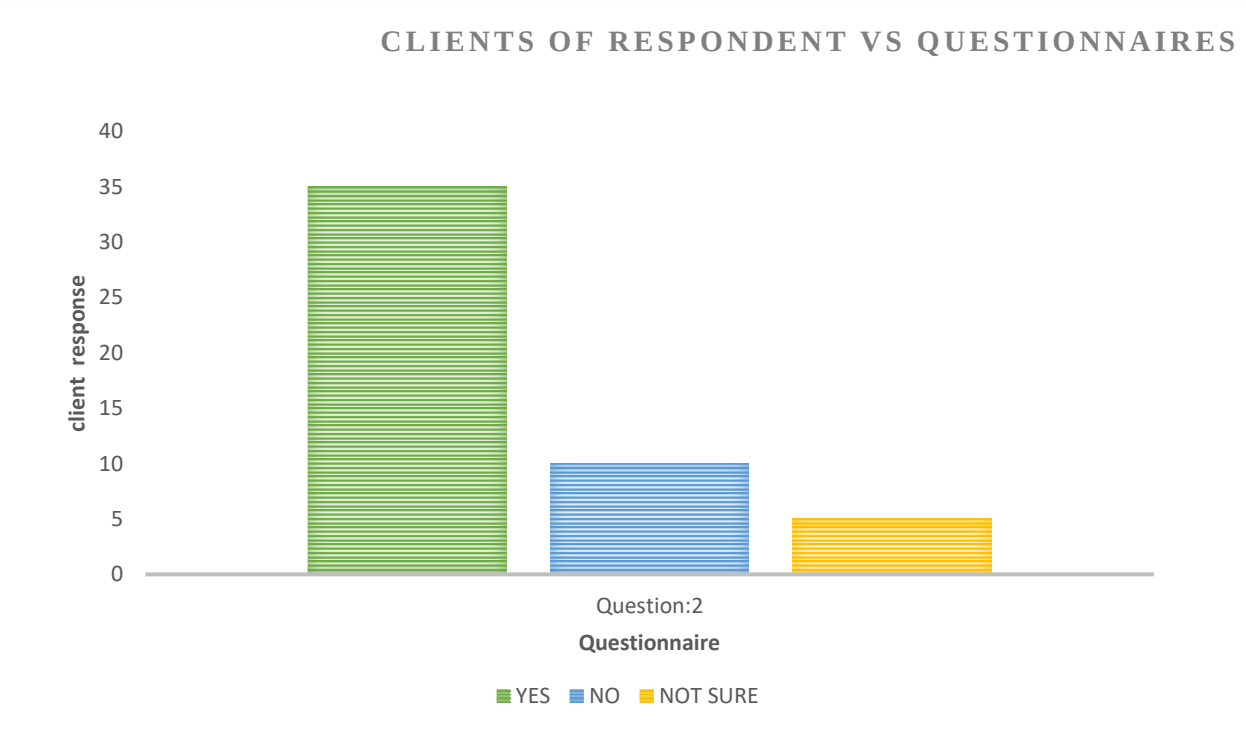
Question1: Gender

a) Male [ ] b) Female [ ]



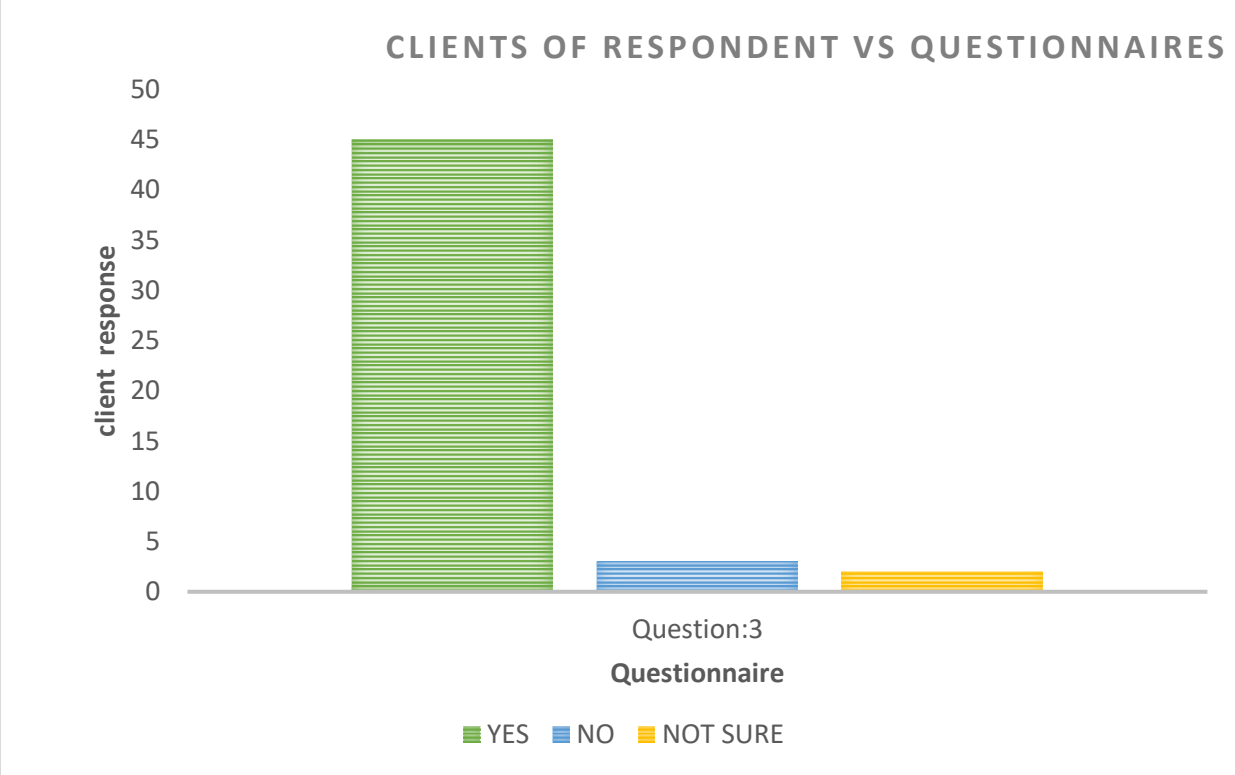
Question2: Do you understood about online system for searching and appointment of advocates?

a) Yes [ ] b) No [ ] c) Not Sure [ ]



Regularity of using rented vehicles the objective of the question was to gather requirements for the project so as to know how much people understand about online system for searching and appointment of advocates. The histogram chart above shows that 35 of respondents know about the system. And 10 of the respondents don't know about online system for searching and appointment of advocates and others 5 staid not sure.

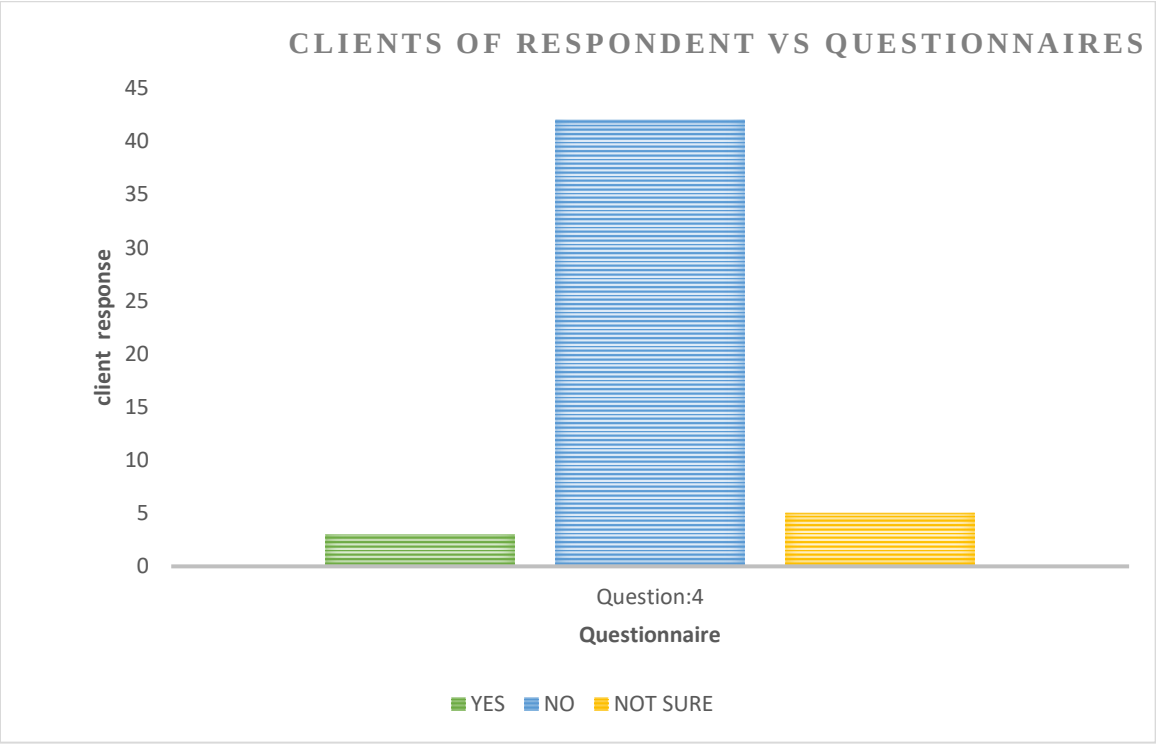
Question3: What challenges do you face with the existing system (Fixed system for searching and appointment of advocates)?



This histogram graph shows that 45 of respondents Yes, 3 of respondents said No and others 2 of respondents said Not Sure, means that they don't know either.

Question 4: Are you satisfied with the existing system (Fixed system for searching and appointment of advocates)?

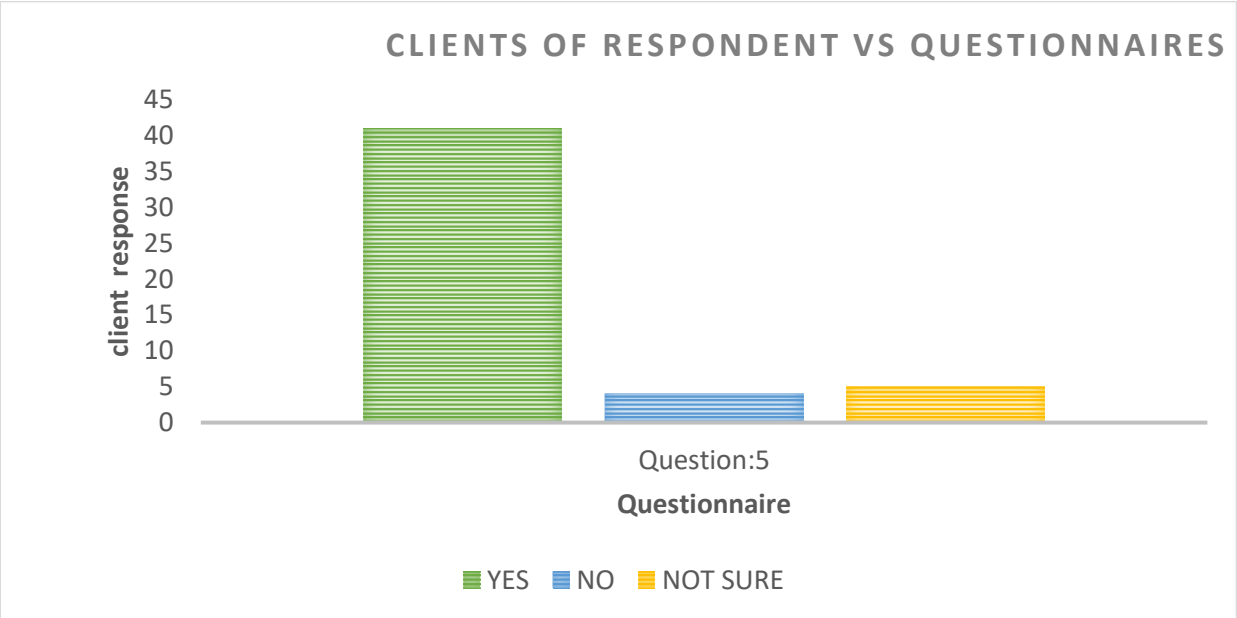
a) Yes [ ] b) No [ ] c) Not Sure [ ]



This histogram graph shows that 3 of respondents Yes, 40 of respondents said No and others 7 of respondents said Not Sure, means that they don't know either.

Question 5: Does a proposed system (online system for searching and appointment of advocates) have a positive impact compared with a current system?

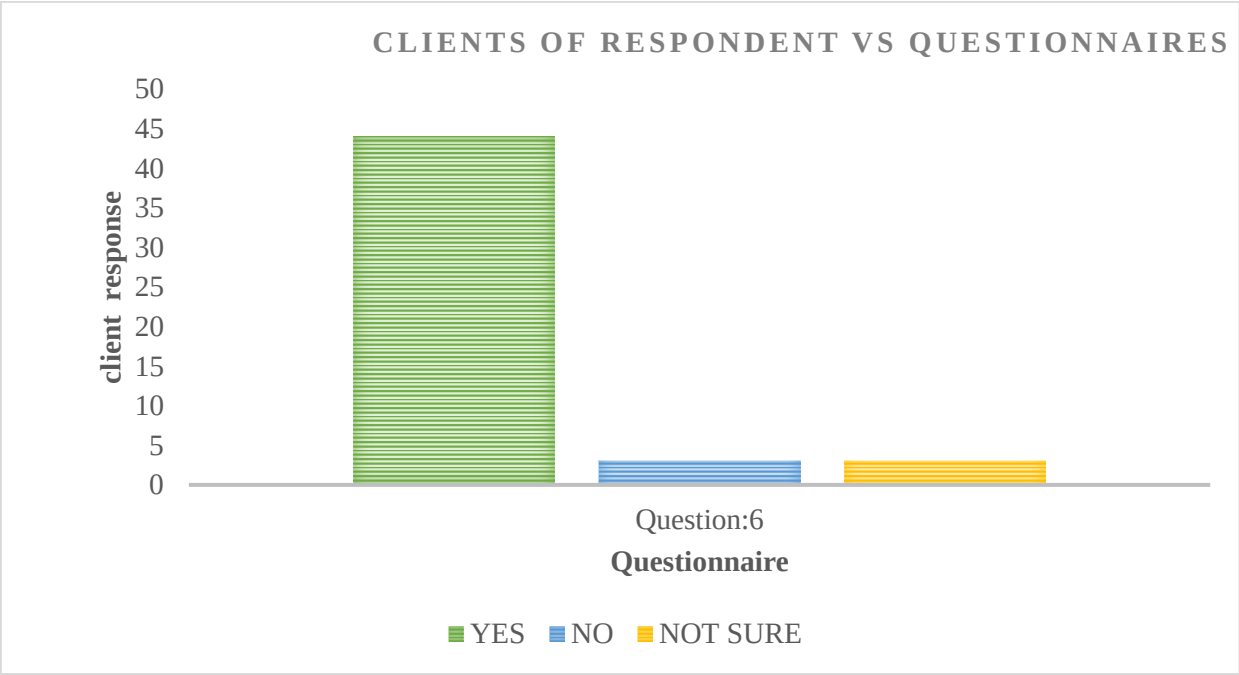
a) Yes [ ] b) No [ ] c) Not Sure [ ]



This histogram graph shows that 40 of respondents Yes, 5 of respondents said No and others 5 of respondents said Not Sure, means that they don't know either.

Question 6: Would you like to use a proposed system (system for searching and appointment of advocates) from the existing system?

a) Yes [ ] b) No [ ] c) Not Sure [ ]



This histogram graph shows that 45 of respondents Yes, 3 of respondents said No and others 3 of respondents said Not Sure, means that they don't know either.

The following where some of the findings from the interview together with the proposed functionalities for the system;

i. Location base – Mbeya
ii. Their existing system is still manual,
iii. The manual system is very unreliable,
iv. They stated clearly, that it is vital for the presence of an automated advocate appointment scheduling system.

And the proposed functionalities for the system were

i. Storage of user data
ii. Registration and keeping records
iii. Feedback from the advocate through normal messages.

3.3 FINDING AND DISCUSTION.

The first objective of the study was to identify and gather requirement for the system for searching and appointment of advocates. The study found that through many people needs the presence of the system for searching and appointment of advocates proposed the requirements for the proposed car rental system. This shows that there is a room for the development system for searching and appointment of advocates which will meet user requirements.

The second objective was to design the system for searching and appointment of advocate's architecture. From the results analyzed above it is stated that there is a need for the development of system for searching and appointment of advocates as many people clearly stated the need for a new system and also proposed the requirements for the development of the system.

The third objective was to implement system for searching and appointment of advocates according to the findings analyzed above them Cleary proves that the system is implementable by manually system which results in difficult during finding and scheduling for an appropriate advocate so this shows that the proposed system can be implemented to the people.

3.3.1 SYSTEM REQUIREMENTS SPECIFICATION

Requirements analysis or requirements engineering is the process of defining user expectations for a System or product. Requirements analysis is an important part of project management.

When we selected this project, we thought about some specific system requirements

- Who will use this system?
- Is it their reliable or not?
- Functional & Non- functional requirements
- Maintenance of the system

3.3.1.1 FUNCTIONAL REQUIREMENT

The functional requirements of the system are like bellow: -

- **Login:** All user, advocate, clients including admin must log in.
- **Registration:** Advocate and client have to register first.
- **System Admin:** System admin can approve advocate registration and can view contact messages that send from application contact us page.
- **Manage Advocate schedule:** Advocate can must manage his regular schedule.
- **View and send massage:** Advocate & Clients can send message and view received messages.
- **View Appointment Request:** Advocate can view who, when and why send him appointment request.
- **Accept, Reject and reply request:** When a client send appointment request its
    - Status will be waiting. Advocate can accept reject or reply on this request.
    - **View Client Profile:** Advocate can view his client's
- **Search Advocate:** anyone search Advocate by address, name and spatiality.
- **View Advocate profile:** Anyone can view Advocate profile and details.
- **View Request Status:** Client can view his request status. And he can delete his waiting and rejected request.
- **Data Validation:** In this stage we have try to validate almost all input field
- **User Interface Design:** It is important to consult the system users and their necessities while designing the use interface.
- **Error Message Design:** The design of error messages is a vital part of the user interface design. We have tried to give proper error message.

3.3.1.2 NON –FUNCTIONAL REQUIREMENT:

The non-functional requirements are given below: -

- **Security Issue:** Database Connection close: After data use, the database connection will close.
- **Data Insert:** Data will be inputted successfully.
- **Data Retrieve:** All data can be retrieved.
- **Upgrade Facility:** It must have upgrade data and system in next time.
- **Availability:** Availability of the Application information.
- **Maintainability:** Data will be maintainable.
- **Quality:** For quality of source code I will be write clean code.
- **User Interface:** Eye set able user interface for this Web Application.
- **Performance:** Performance of data speed will be fast.
- **Usability:** Nice data illustration.
- **Reliability:** Reliable of data.
- **Accuracy:** Data Representation.

3.3.1.3 STAKEHOLDERS

Internal Stakeholder

- Administrator

External Stakeholder

- Advocates
- Clients

3.4 PROJECT SCHEDULE

In this stage in want to show our project Gantt chart.

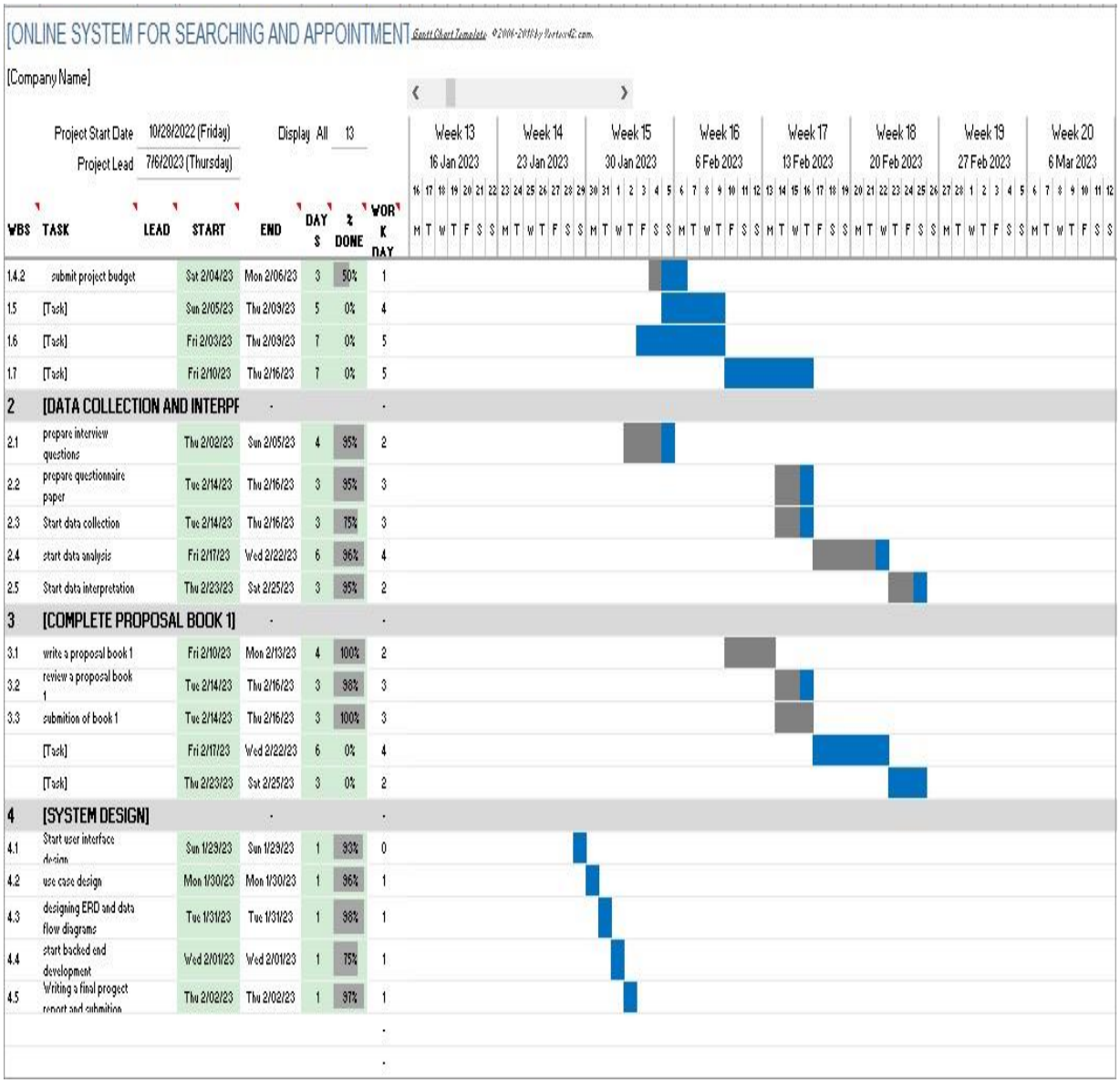


Figure 3.4 project Gantt chart

CHAPTER FOUR

SYSTEM DESIGN

4.1 Introduction

Design is the first step in the development phase for any techniques. Software design is a process which requirements are converted into a representation of system.

Once the System requirements have been analyzed and specified the software design involves three technical activities - design, coding, implementation and testing that are required to build and verify any application or system.

The design activities are of main importance in this phase, because for this activity depends implementation, testing and future development.

There are various kinds of methods in software design I will show some of this for my application: -

- Use case Diagram
- Sequence Diagram
- Flowchart
- Data Flow Diagram
- Entity Relationship Diagram
- Activity Diagram

4.1.1 USE CASE DIAGRAM

The main purpose of use case diagram is to capture the dynamic part of a system. But this definition is too standard to describe the purpose.

- The purpose is to show the connections between the use case and actor.
- To symbolize the system requirements from user's
- An actor can be the end-user of the system or an external system.

4.1.1.1 Clients use case:

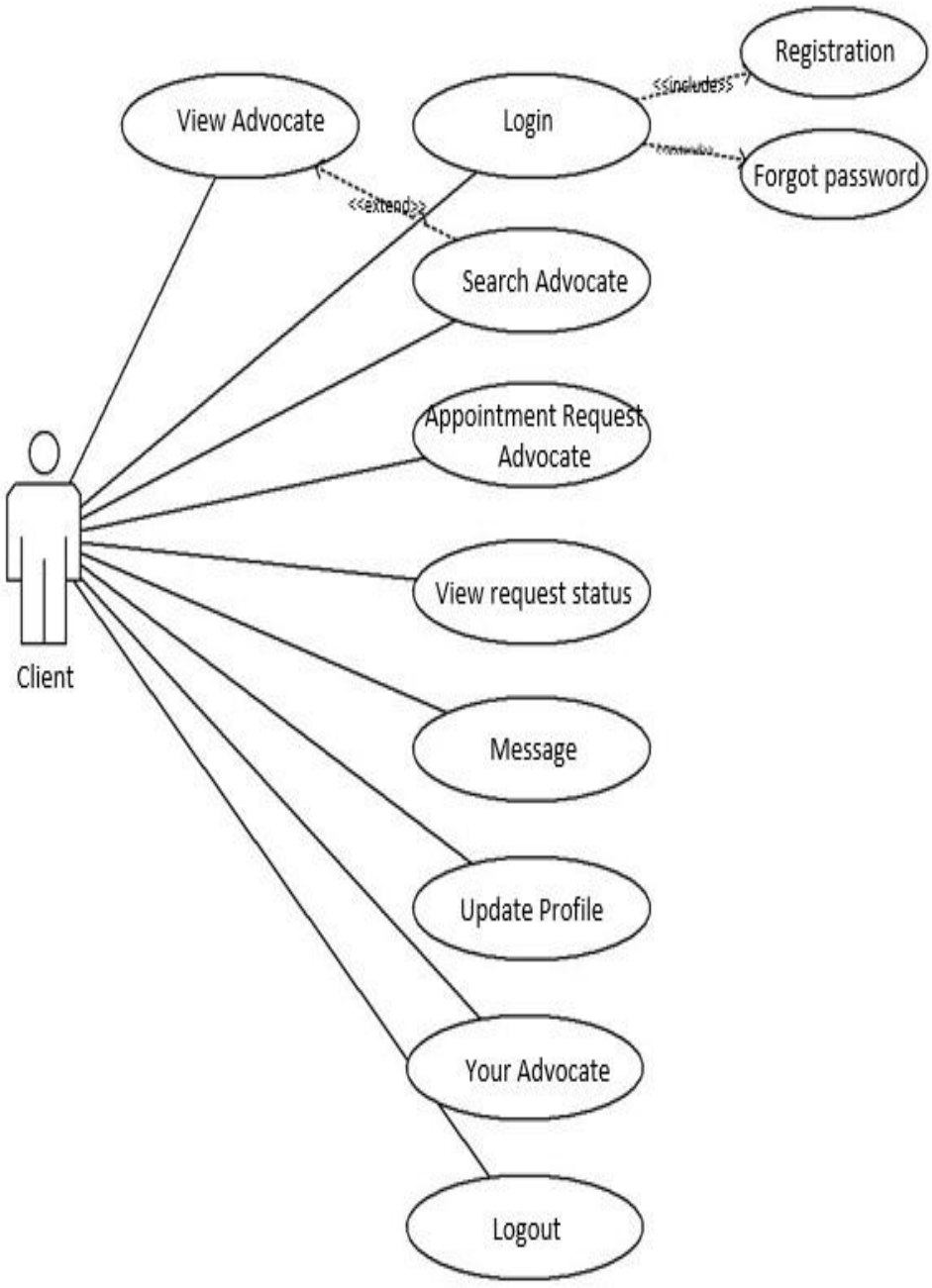


Figure4.0.1use case diagram for client module

4.1.1.2 Advocate Use Case:

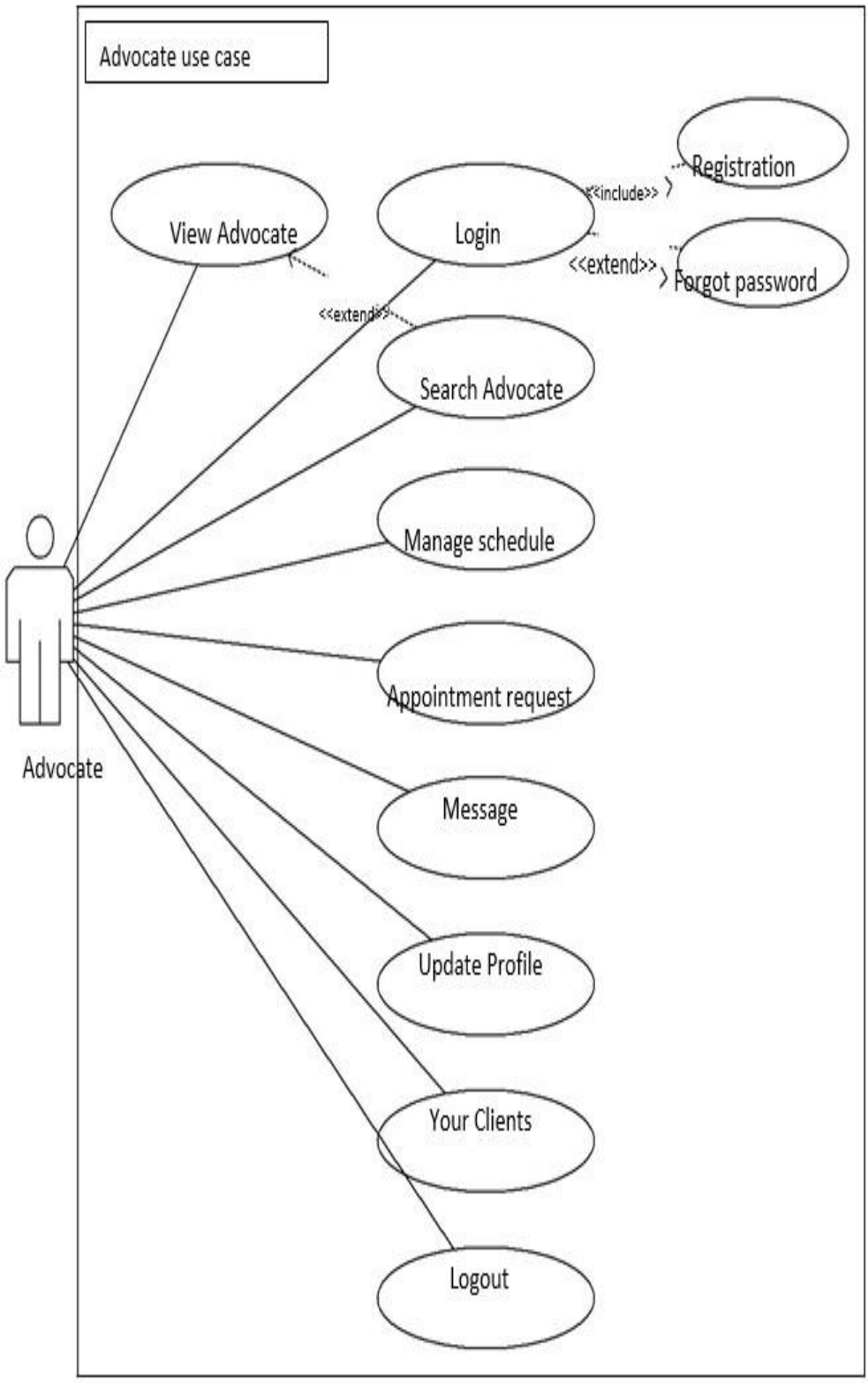


Figure 4.0.2use case diagram for Advocate module

4.1.1.3Admin Module



Figure 4.0.3use case diagram for Admin module

4.3 SEQUENCE DIAGRAM

Sequence diagram and collaboration diagram are called INTERACTION DIAGRAMS. The purposes of interaction diagrams are to visualize the collaborative behavior of the system. Now I am show sequence diagram for my application

4.3.1 Client Sequence Diagram

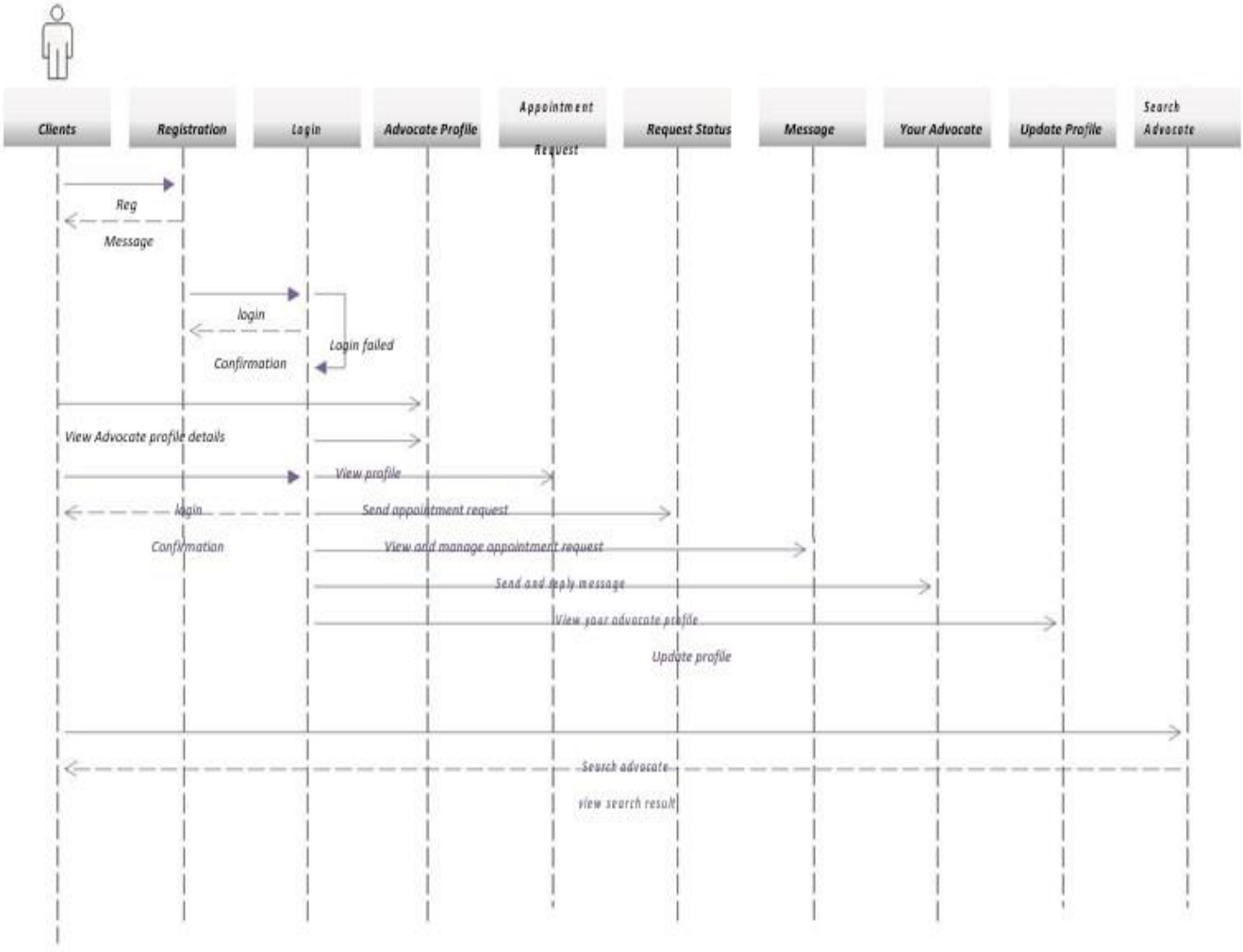


Figure4.0.4 Client sequence diagram

4.3.2 Advocate Sequence Diagram

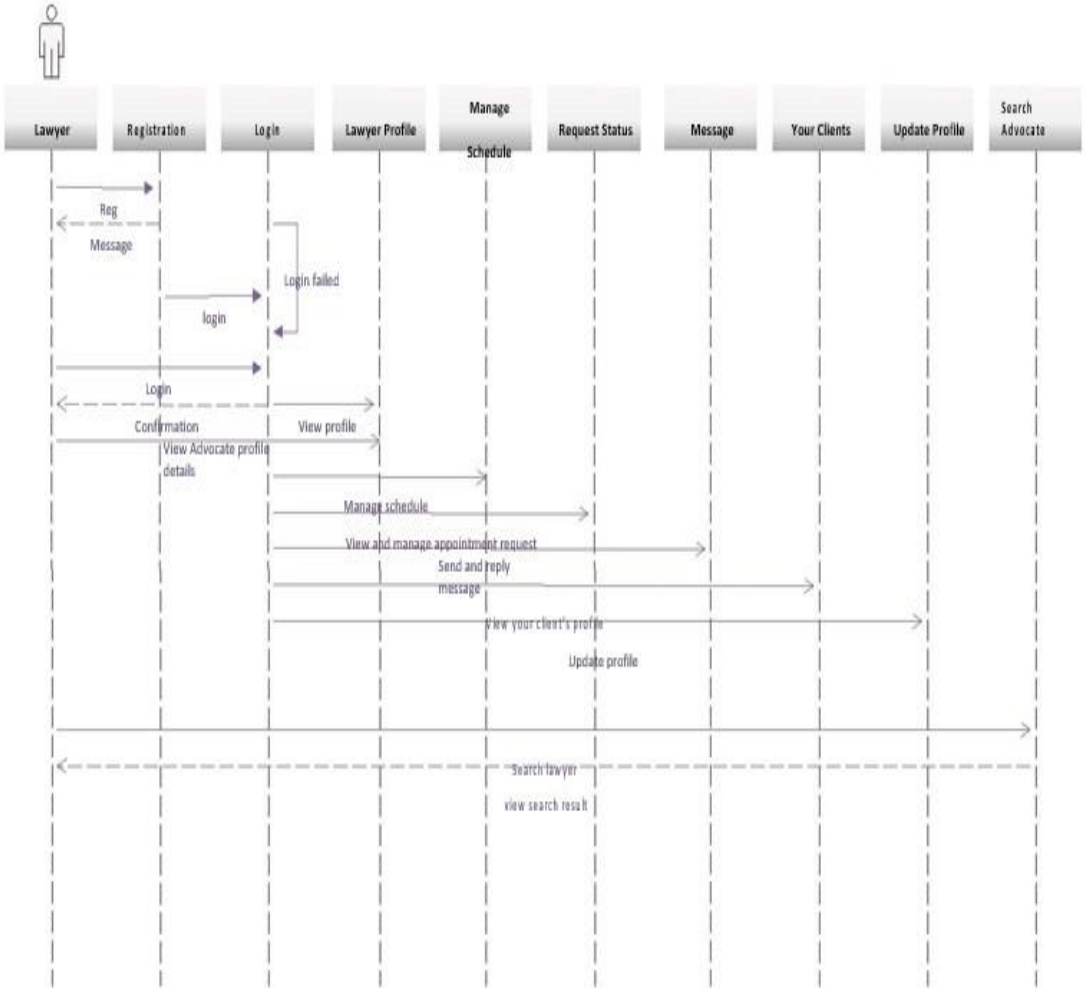


Figure 4.0.5 advocate sequence diagram

4.3.3 Admin Sequence Diagram

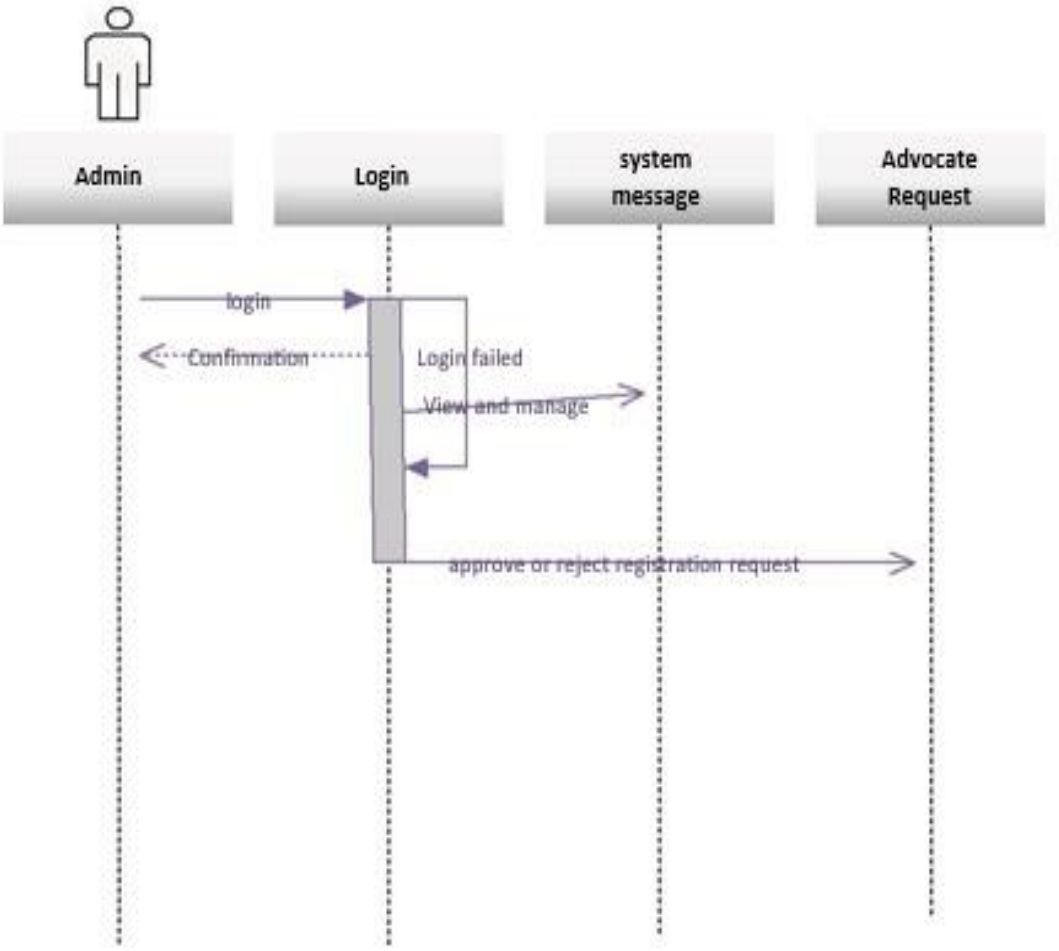


Figure 4.0.6 Admin sequence diagram

4.4 FLOWCHART

Flowchart is a diagram that describe flow of a system. Here I want to try describe the flow of my application.

4.4.1 Flowchart for clients

Here describe what can do a client by this system.

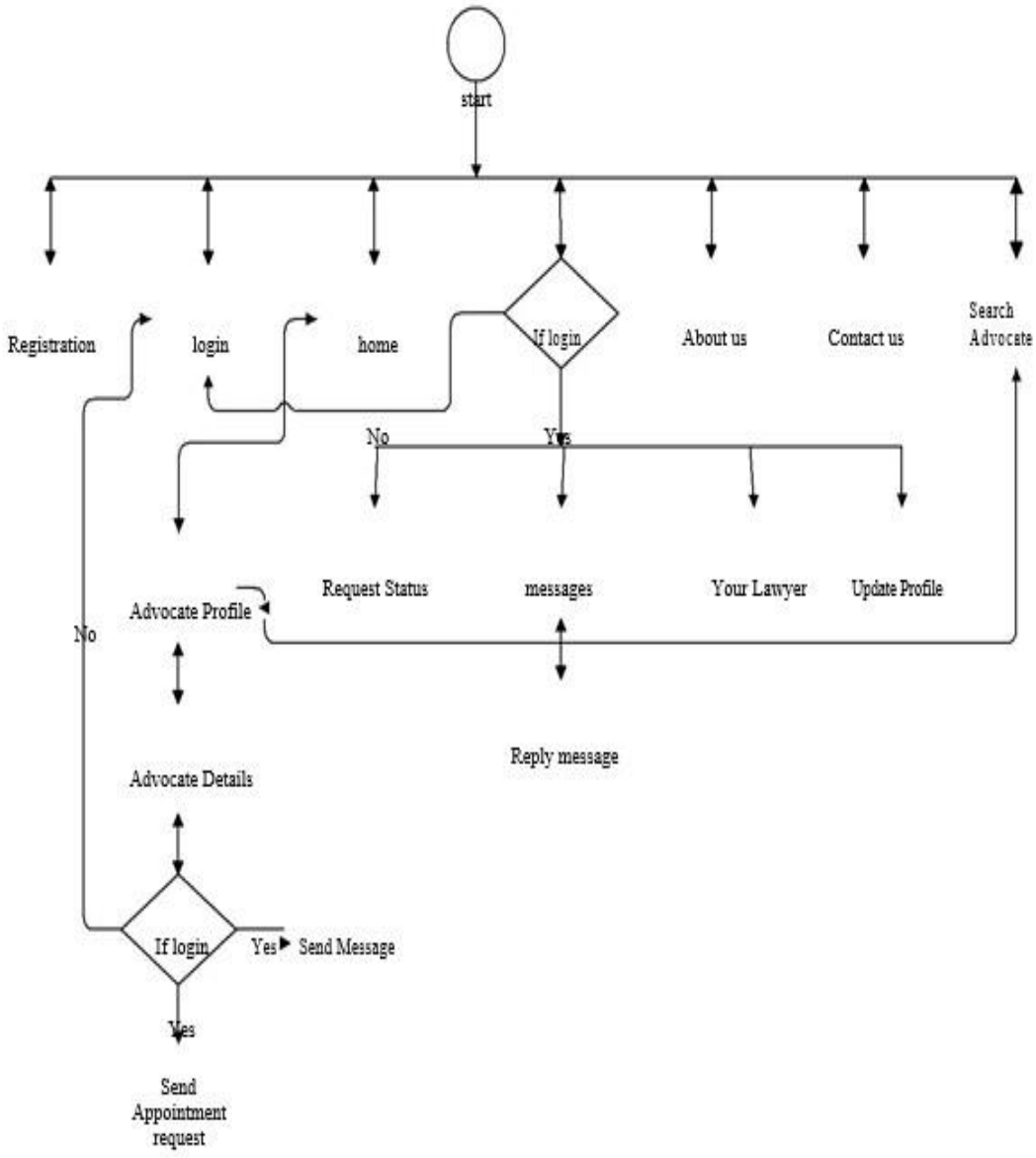


Figure4.0.7 Flowchart for client's module

4.4.2 Flowchart for Advocate system

Here I want to describe what can do an advocate by this system.

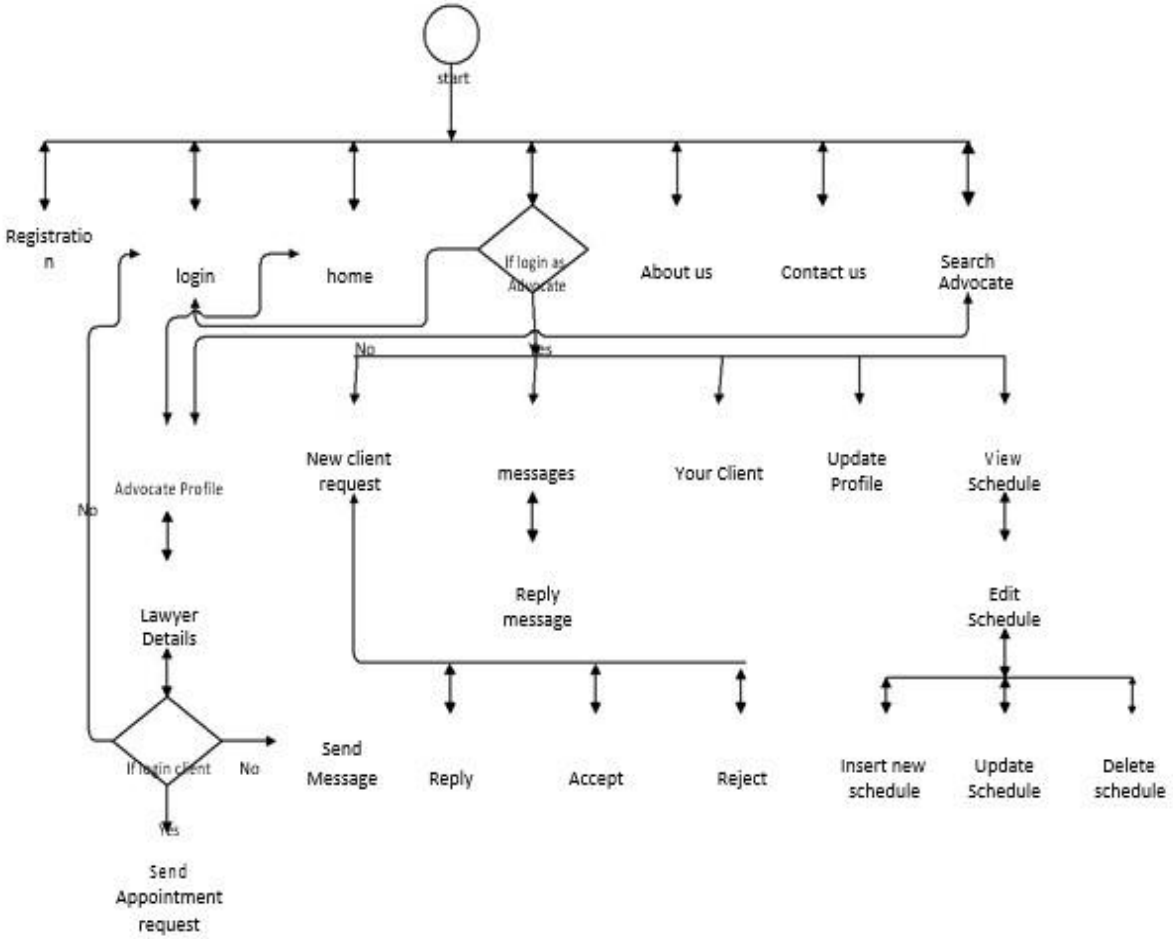


Figure 4.0.8 Flowchart for Advocate's module

4.4.3 Flowchart for admin

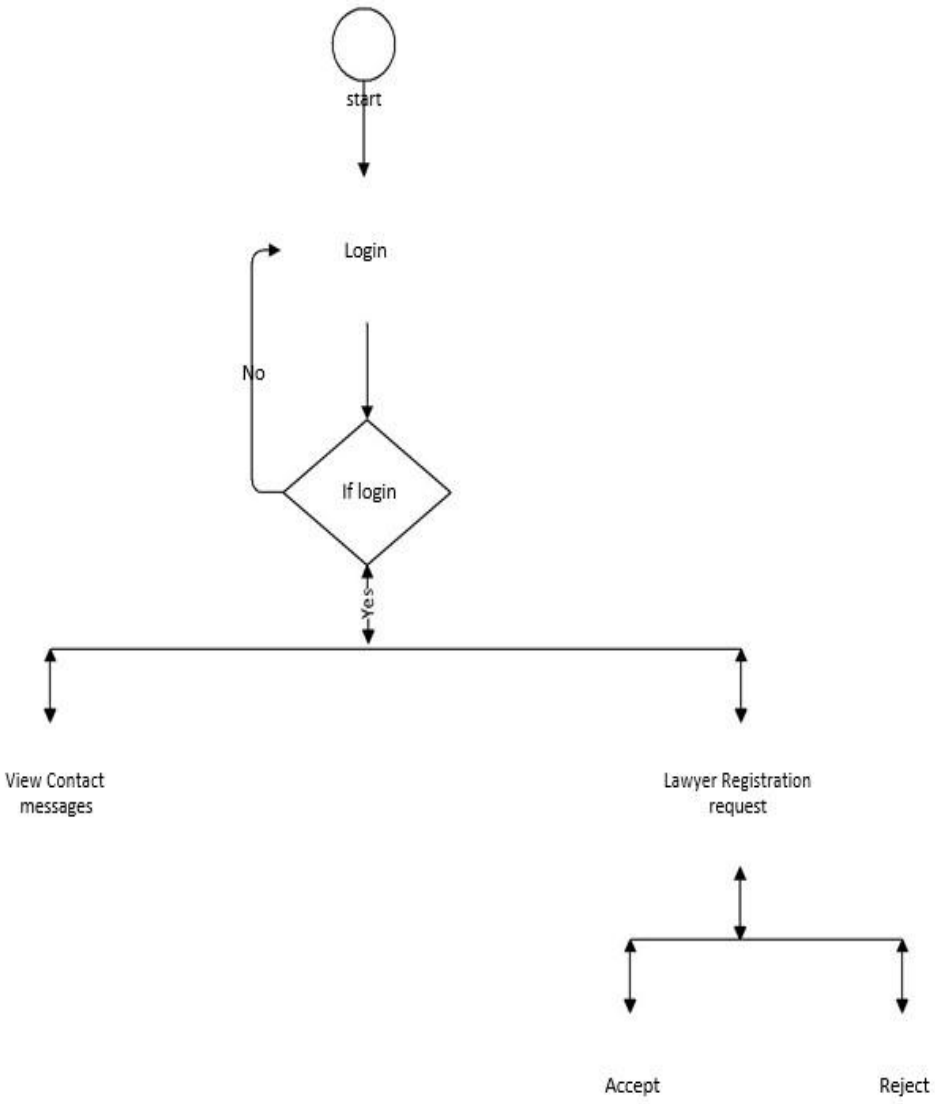


Figure 4.0.9 Flowchart for Admin module

4.4.4 DATA FLOW DIAGRAM (DFD)

DFD is a graphical illustration of the flow of data over an information system, modelling its process characteristics. A DFD is often used as an initial step to create an overview of the system. Data objects symbolized by labeled arrows and transformation are symbolized by circles also called as bubbles.

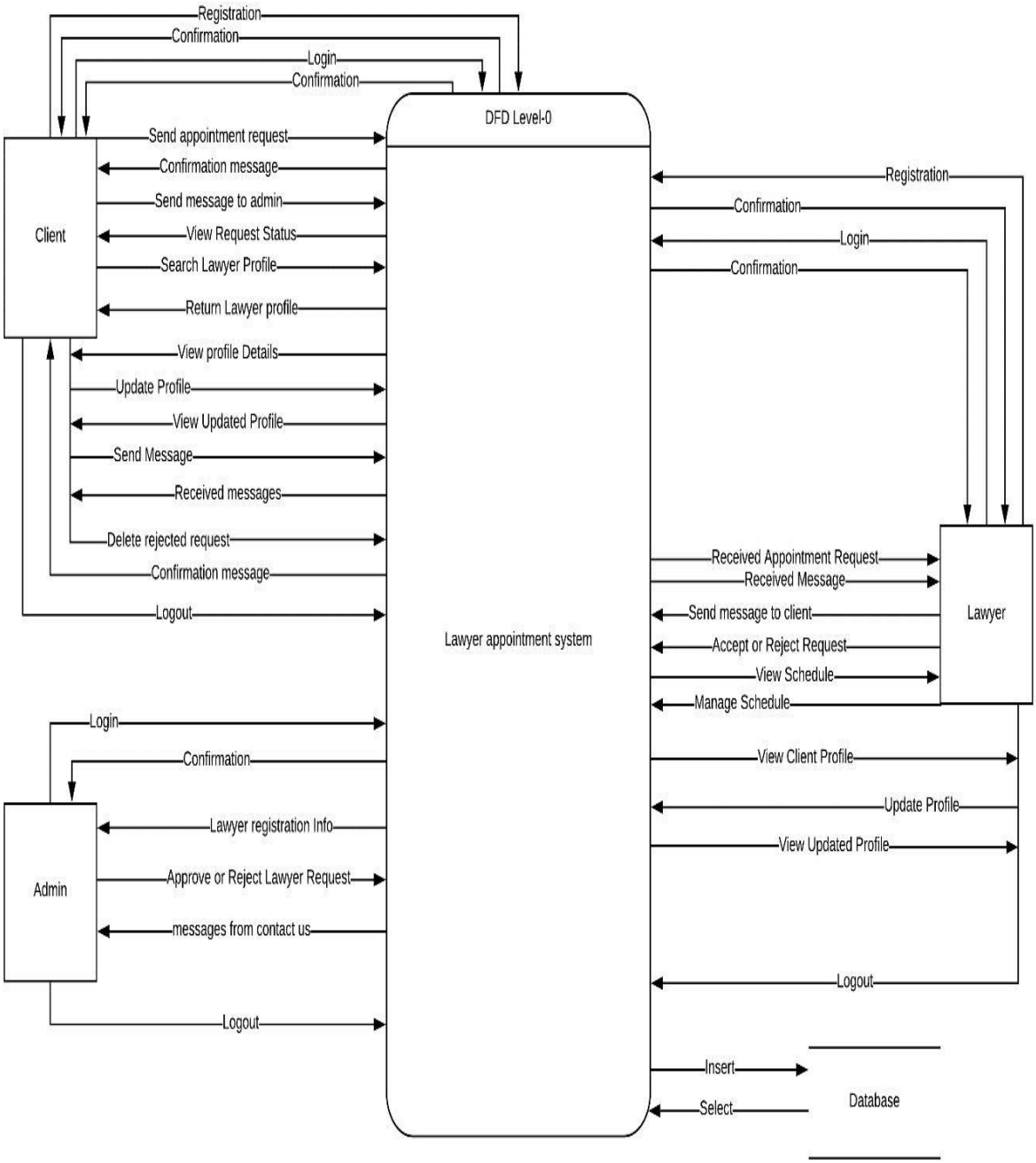


Figure 0.10 Data Flow Diagram (DFD) level 0

4.4.5 ENTITY RELATIONSHIP DIAGRAM (ERD)

ER-Diagram is a graphically representation of entity and relationship of entity. There are two types of ERD- Logical and physical diagram. In this stage I want present my system physical ERD.

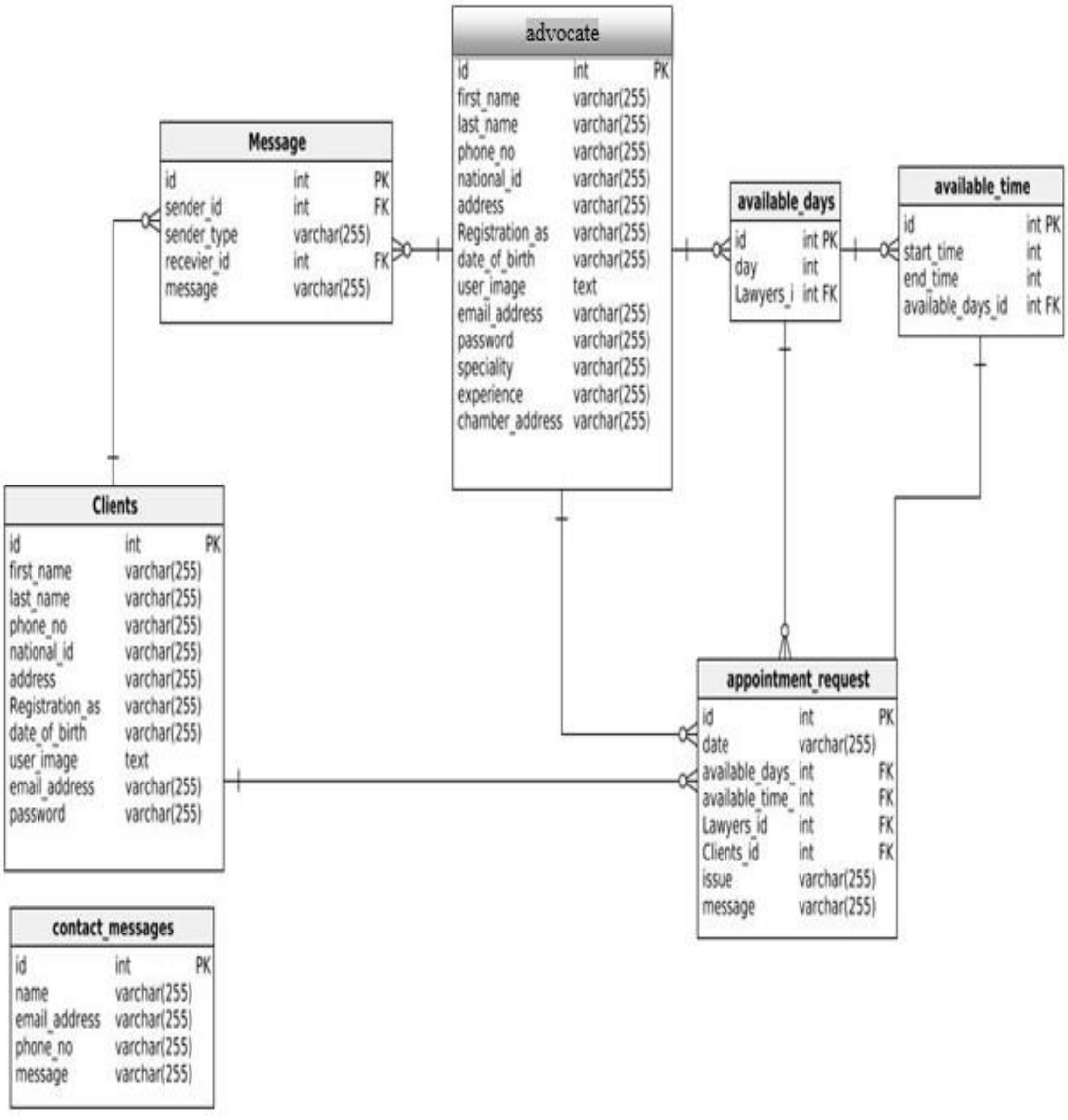


Figure 4.0.11 Entity Relationship Diagram (ERD)

4.5 ACTIVITY DIAGRAM

The activity diagram of a system describes the processing flow within each operation.

Now we show activity diagram of **advocate appointment system.**

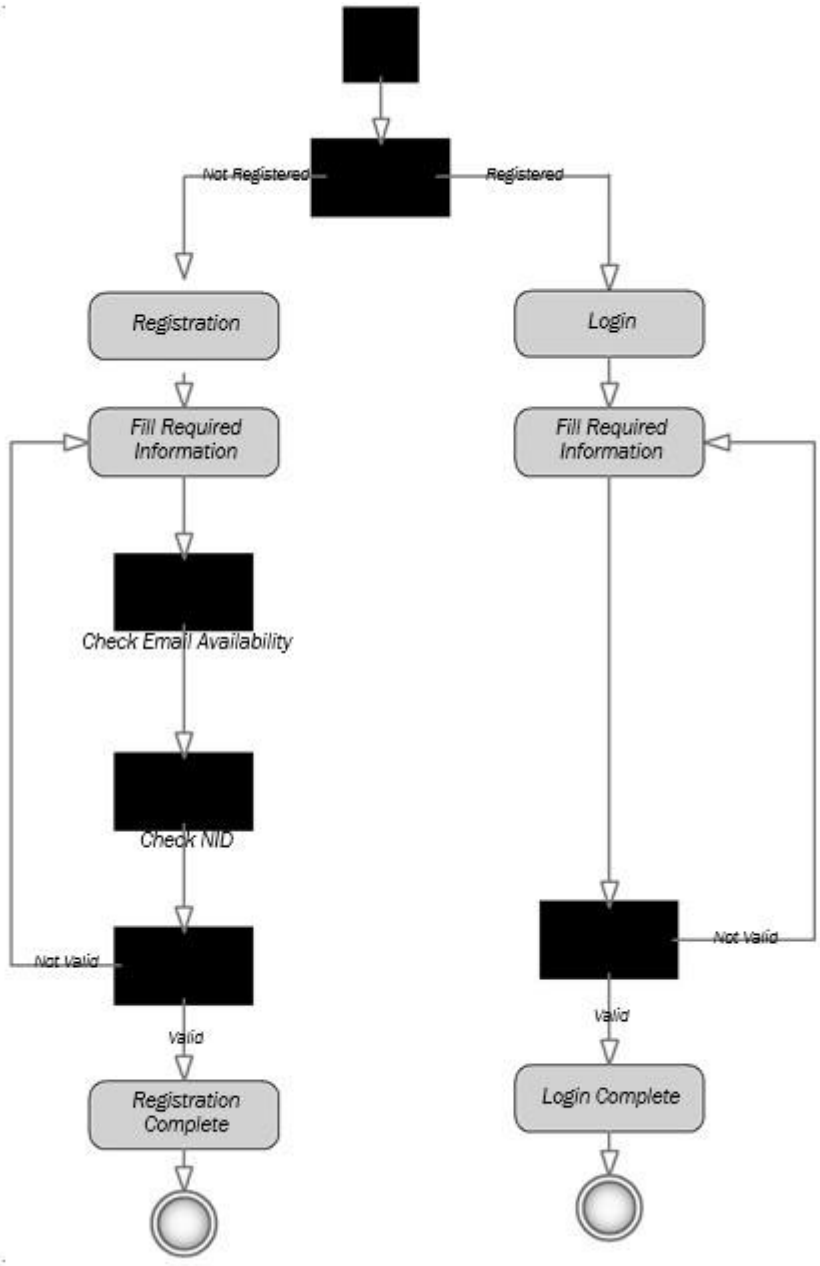


Figure4.0.12 Activity Diagram for registration and login

4.5.1 Send appointment request activity diagram

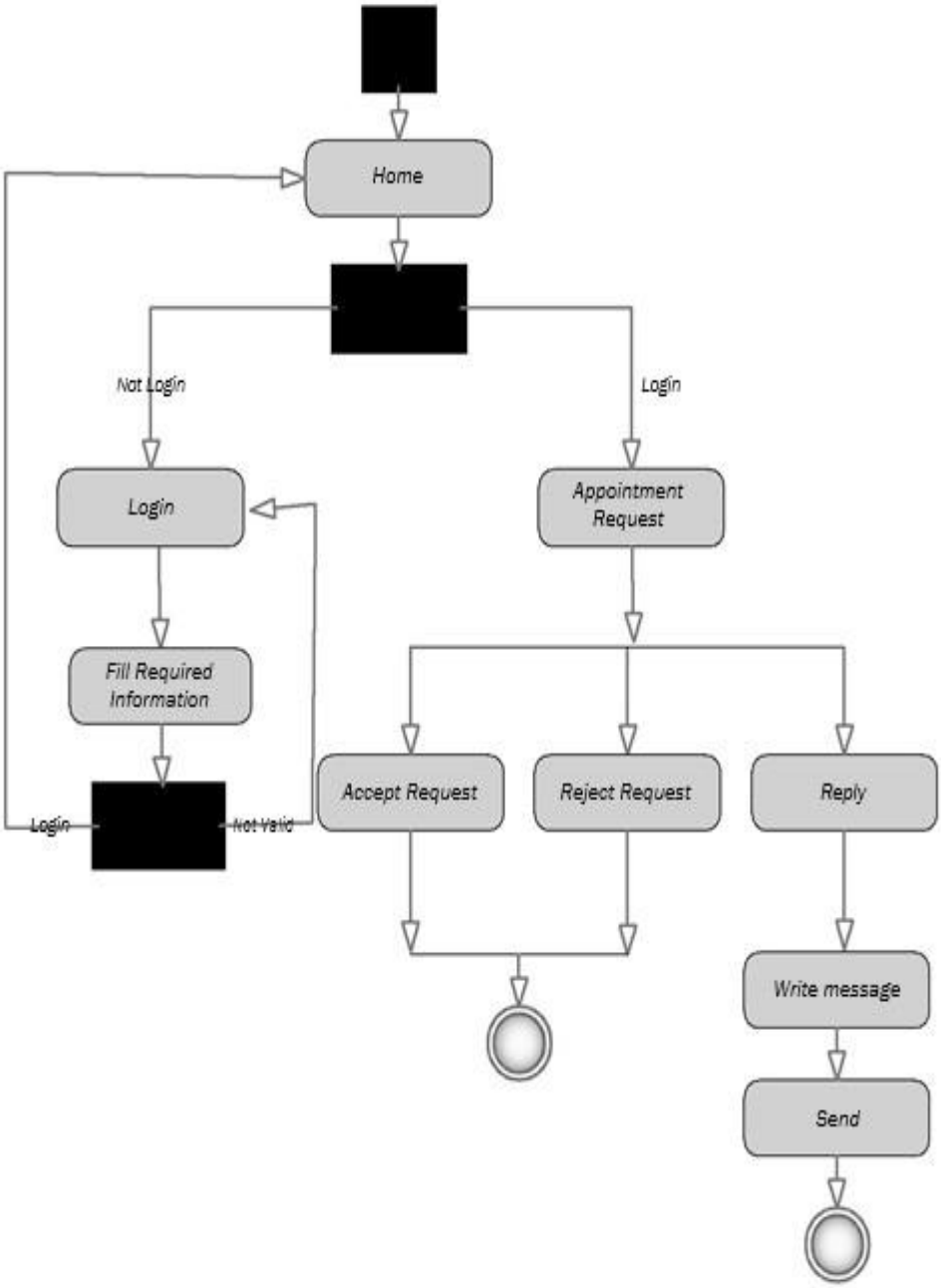


Figure 4.0.13 Activity Diagram for sending Appointment

4.5.2 Activity diagram for manage appointment request

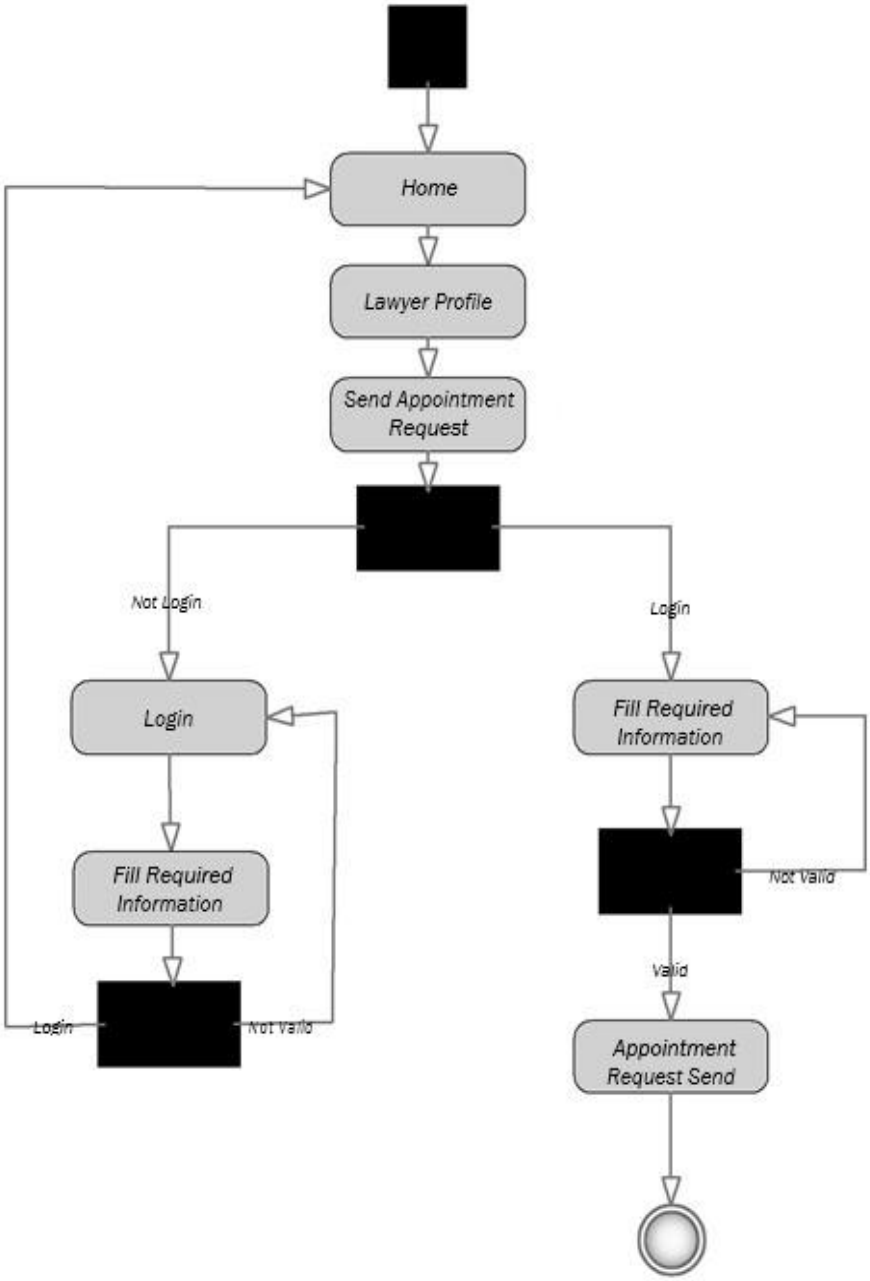


Figure 4.0.14 manage appointment request Activity Diagram

4.5.3 Update profile activity diagram

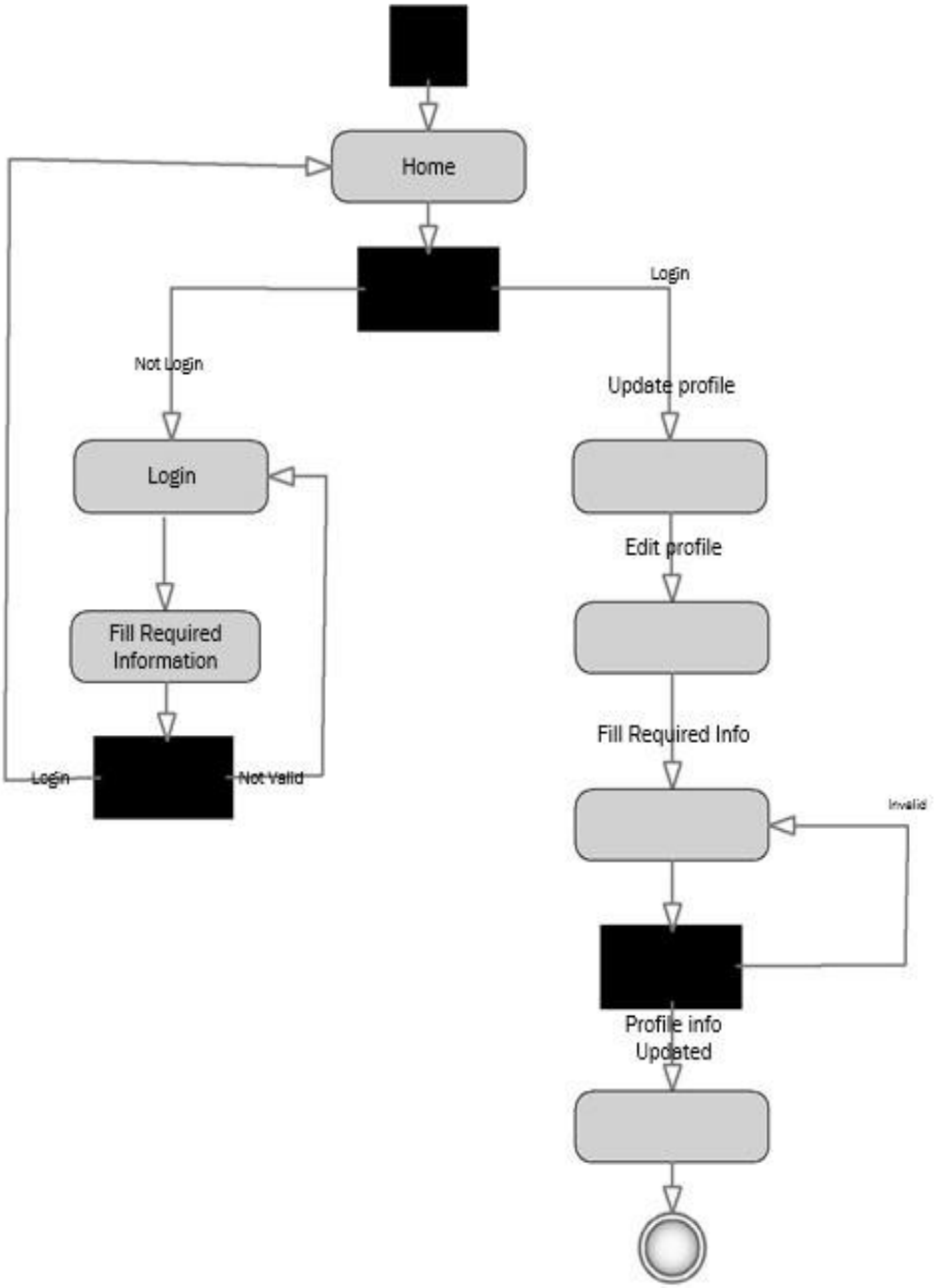


Figure 4.0.15 Update profile information Activity Diagram

4.6 IMPLEMENTATON

The system design is mapped. Coding and all the logics and functions of the business processes of the system are implemented with appropriate codes. User friendly and descriptive graphical user interface is made with illustrative indication of the features. The description of the Graphical User Interface (GUI) and features are followed in the next section. Along with safe operations, proper authentication schemas are implemented to protect the system from unauthorized access and ensure the validity and confidentiality of the data in the system. The object designs and data models are implemented. Database is created in MySQL. HTML pages and forms are implemented for user interaction purposes.

4.6.1 HARDWARE & SOFTWARE SPECIFICATIONS

This stage we want to supplication describe what's need

4.6.1.1 Hardware Requirements:

- PROCESSOR: Dual Core or above
- RAM: 2GB or above
- Cache Memory: 2MB or above
- HDD: 20GB or above

4.6.1.2 Software Requirements:

- Php Storm, PHP Designer, Notepad++, Sublime Text, Code lobster PHP Edition etc.
- Database: MySQL
- Web-Server: Apache

4.6.2 UI IMPLEMENTATION

4.6.2.1 Home page

Now I am showing home page of my application. In home page anyone can view Advocate profiles.

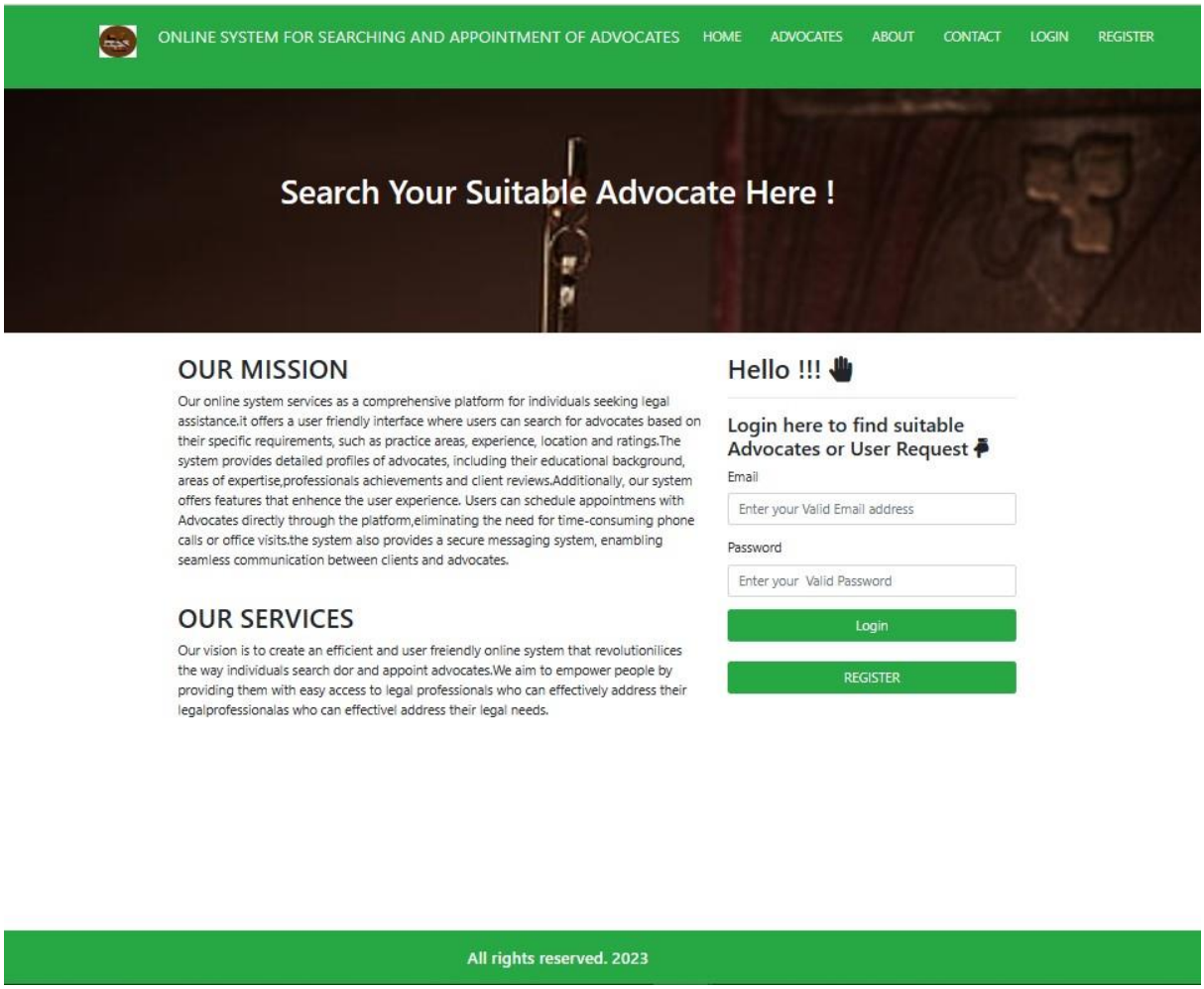


Figure 4.0.16 system home page

4.6.2.2 Registration page

In this stage user can registration as an advocate or client. User must have to fill up all the field other ways it shows error message.

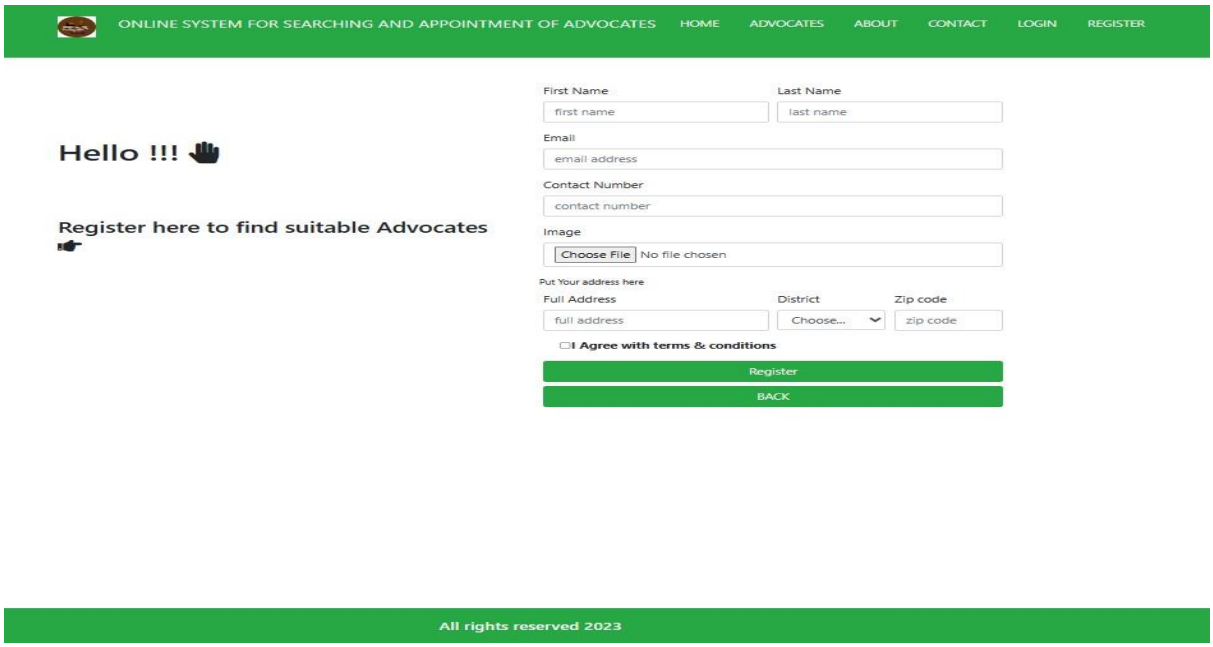


Figure 4.0.17 User Registration page (part 1)

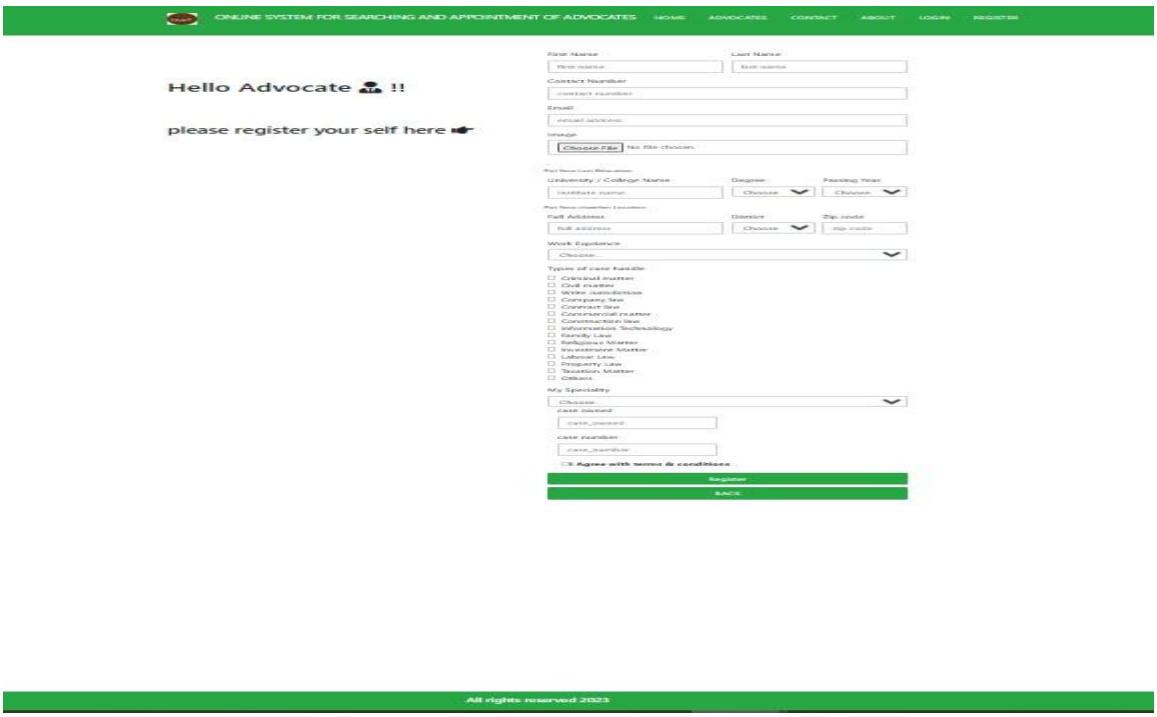


Figure 4.0.18 Advocate Registration page (part 2)

4.6.2.3 Login page

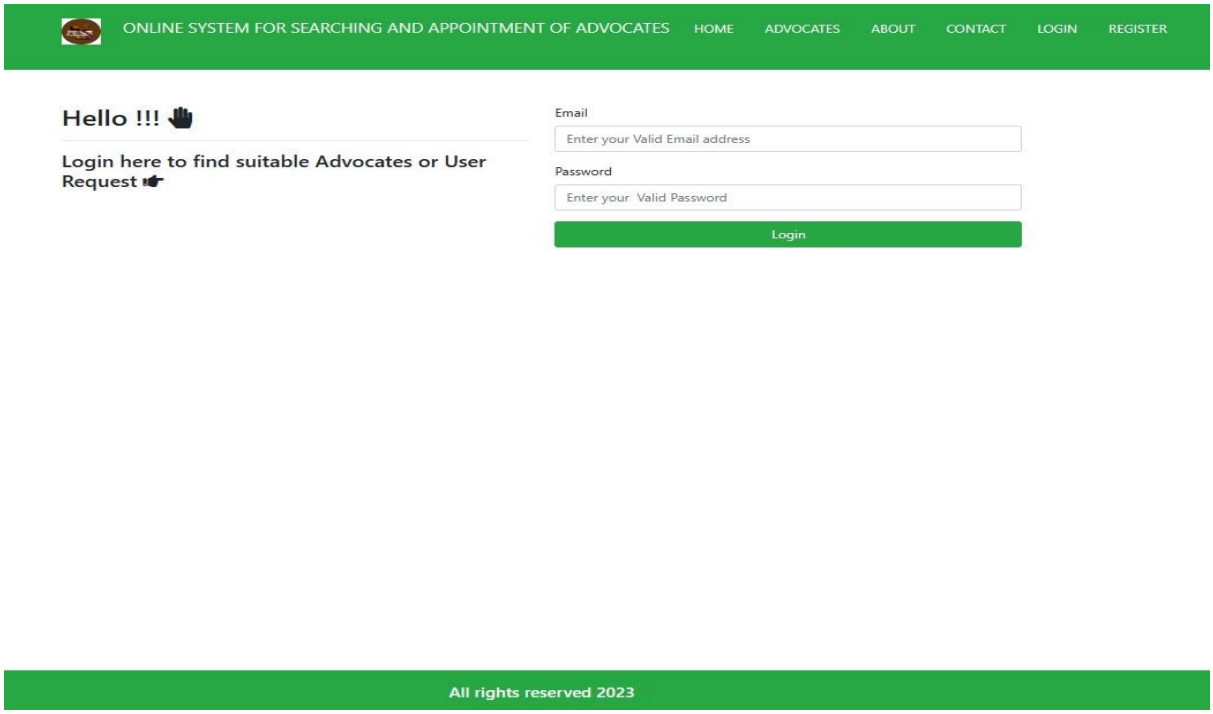


Figure 4.0.19 User Login page

4.6.2.4 Update Profile:

User can update his profile from this page.

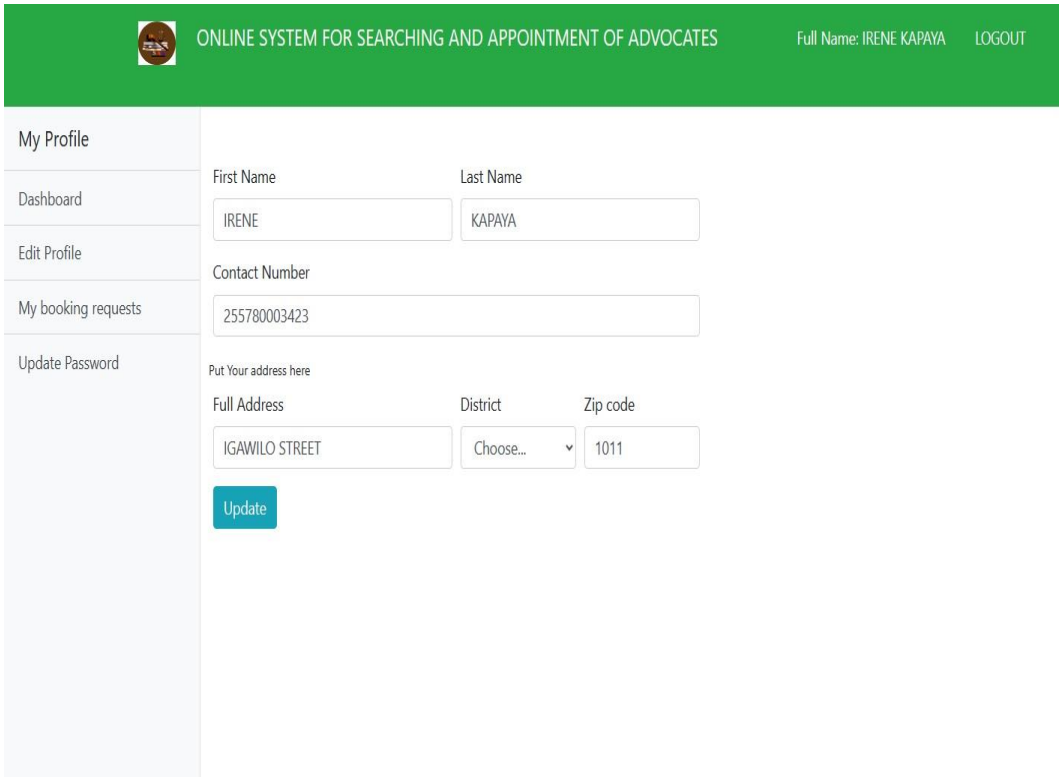


Figure 4.0.20 View Profile Detail update page

4.6.2.5 Search advocates page

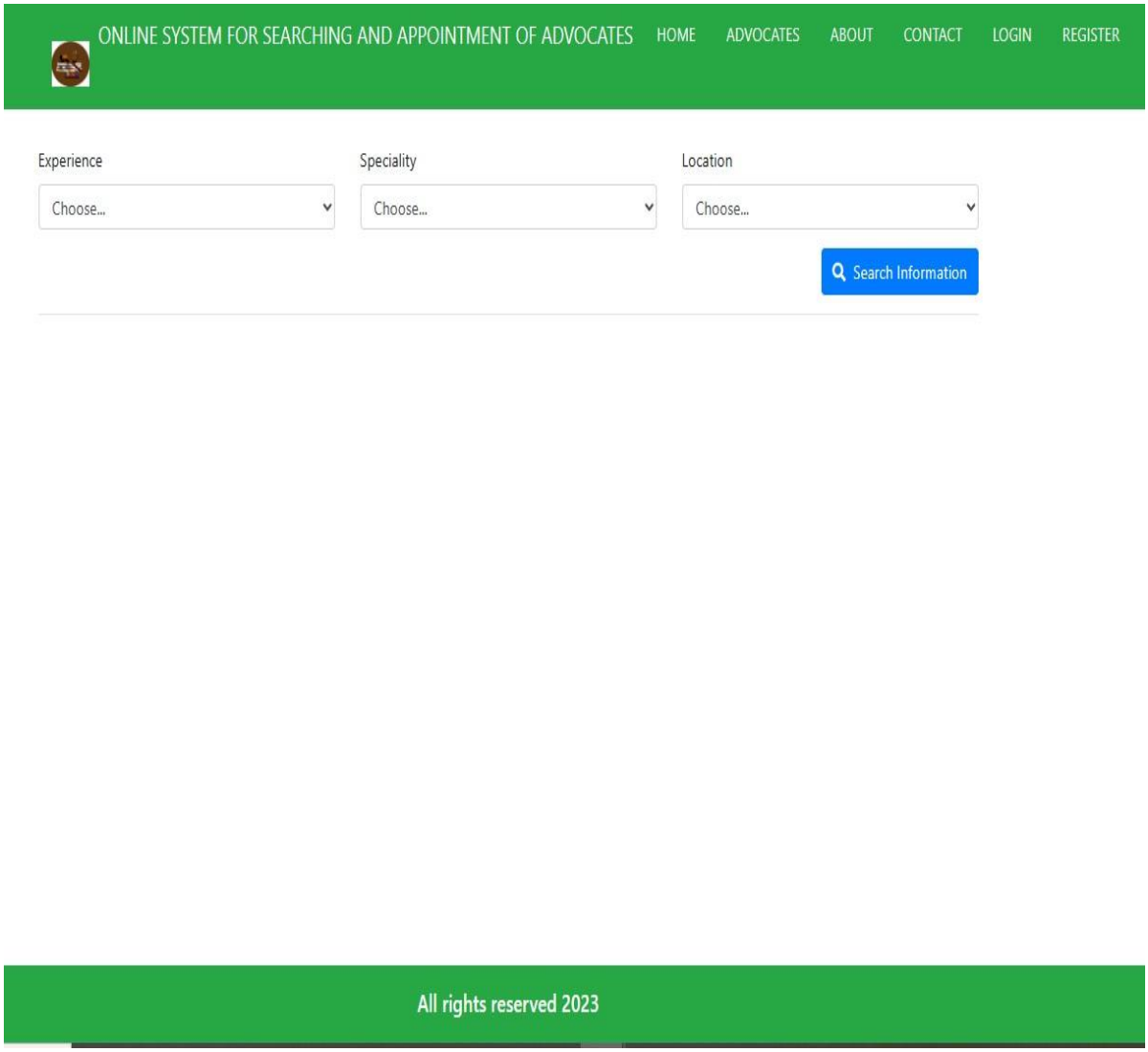


Figure 4.0.21 search advocates information page

4.6.2.5 Appointment Page for advocate:

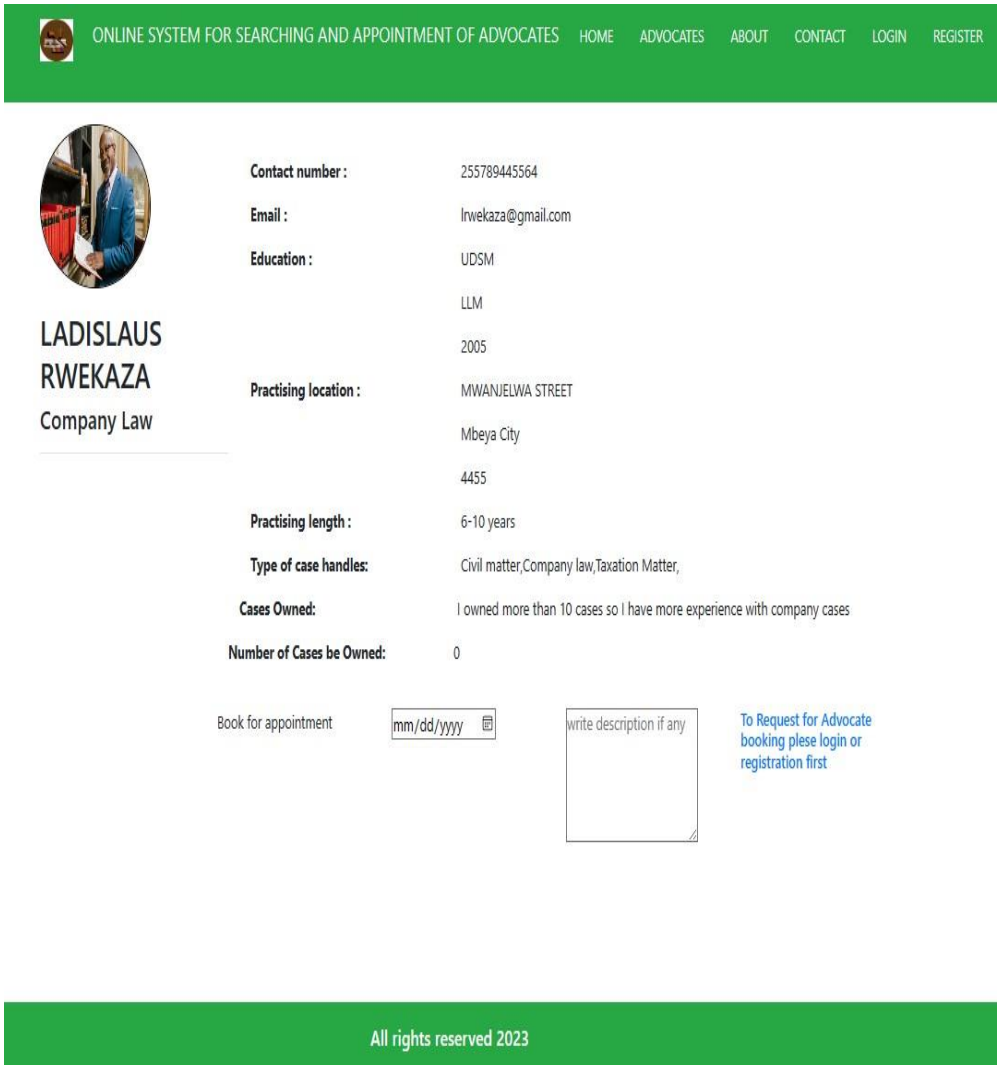


Figure 4.0.22Advocate Appointment page

4.6.2.6 Advocates profile lists

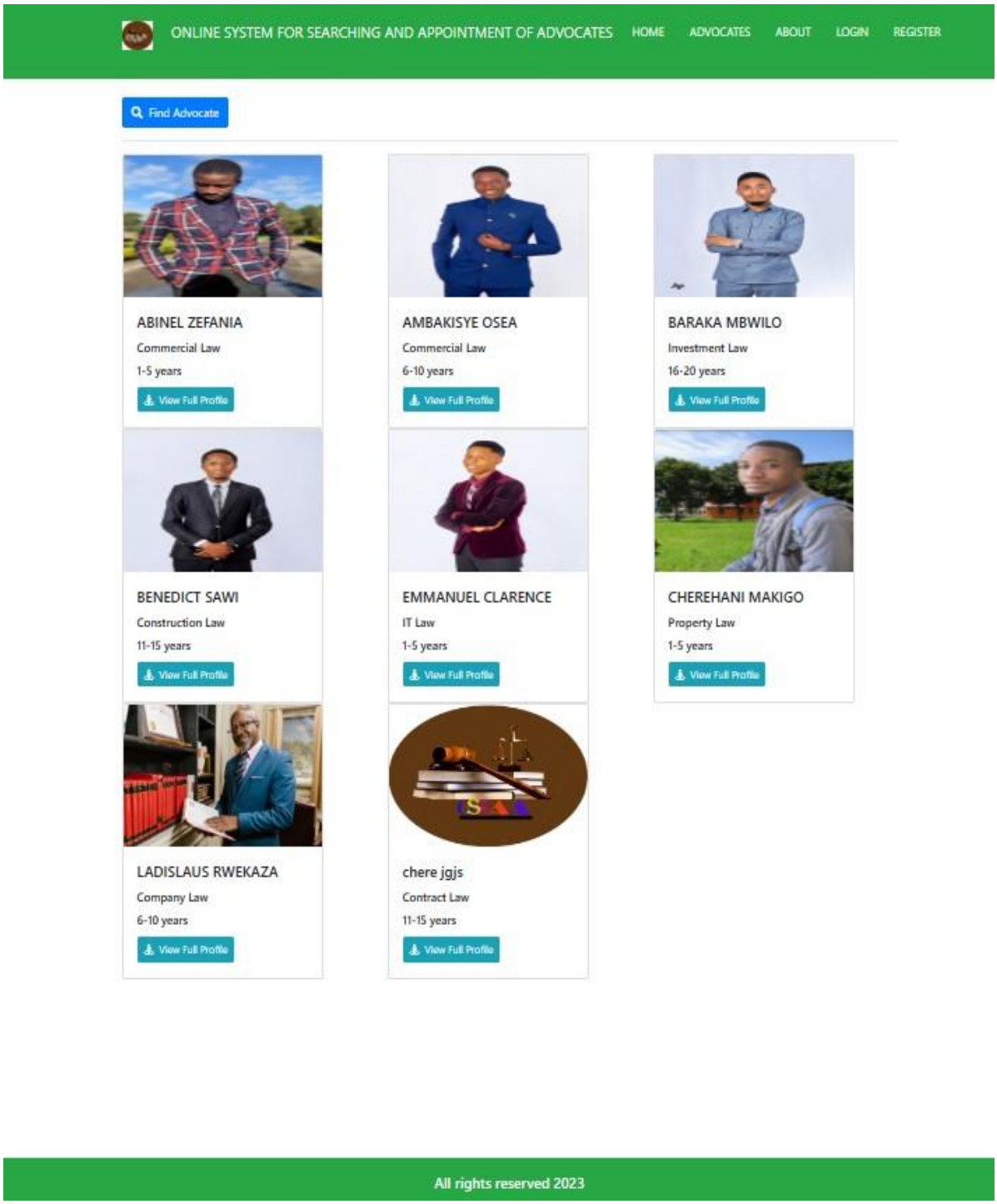


Figure 4. 0.23 Advocates profile

4.6.2.7 Send Appointment Request:

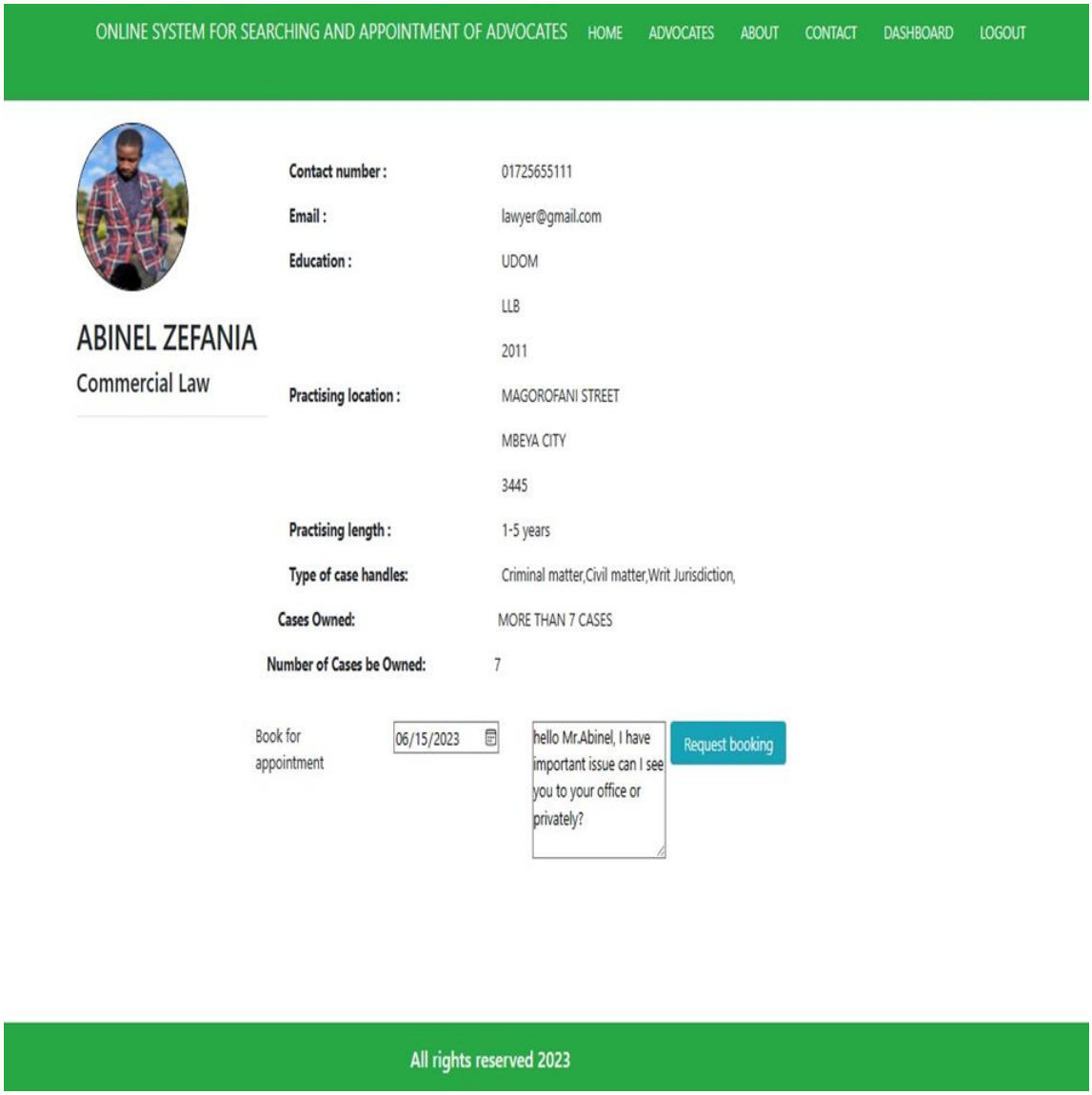


Figure 4.0.24 Client send Appointment Request page for advocate

4.6.2.8 Contact Us page

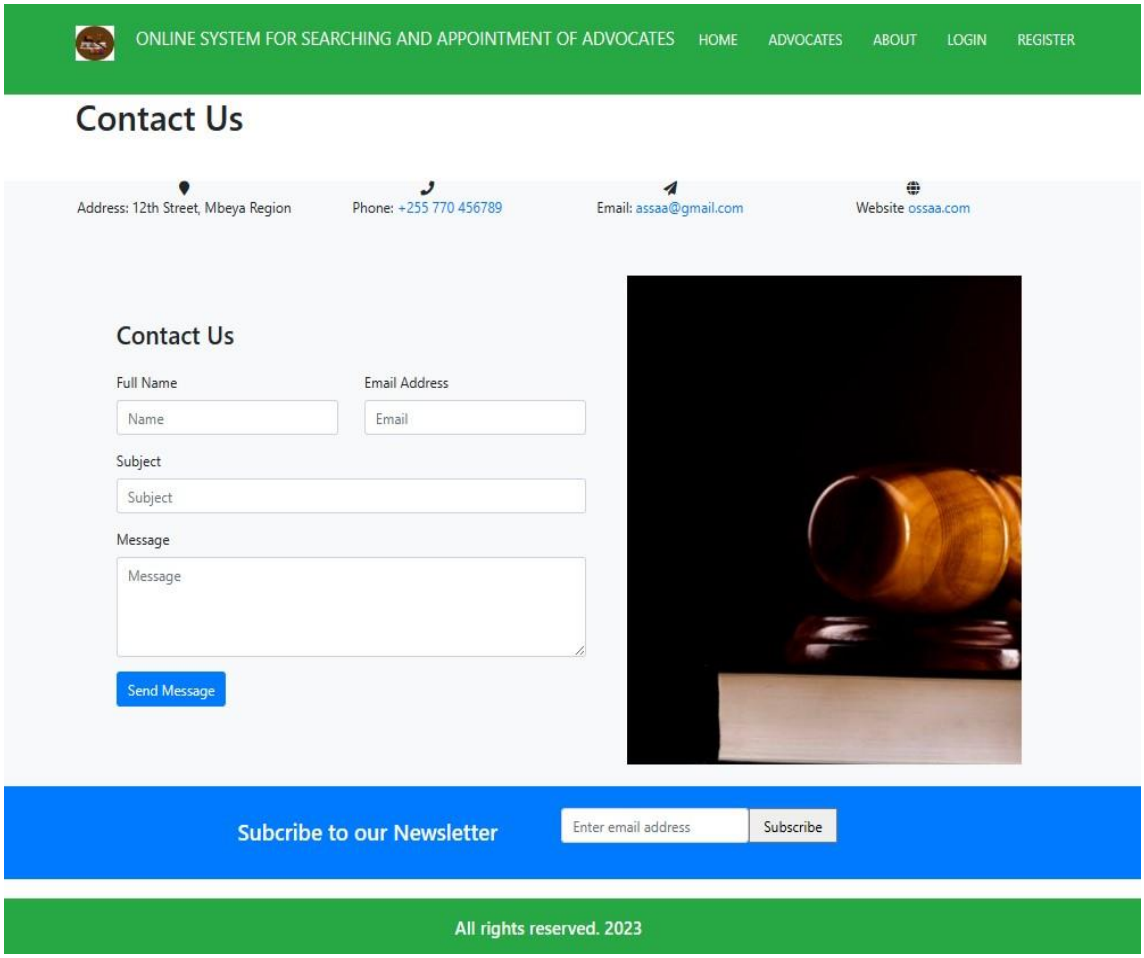


Figure 4.0.25 Contact Us page

4.6.2.9 About Us page



Figure 4.0.26 About us page

CHAPTER FIVE

RECOMMENDATION AND CONCLUSION

5.1 Recommendation

This application avoids the manual work and the problems concern with it. Well I have worked hard in order to present the website. Still, I found out that the project can be done in a better way. Due to time and technology concern I am unable to deploy the online payment system and notification system and other small features. For that reason, I am going to develop new features like following:

- Online Payment System using payment gateways
- Notification System
- View laws and search laws.
- Review of Advocates from clients.

5.2 Conclusion

The application is designed in such a way that future modifications can be easily done. The following conclusion can be assumed from the development of the project. Automation of the entire application improves the good organization. It delivers a friendly graphical UI and gives proper access to the authorized users depending on their authorizations. It effectively overcomes the delay in communications. Updating of data becomes very easier. Application security, data security and reliability are the striking features. The System has passable scope for modification in future if it is essential

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PROJECT COST SHEET

S/NO	ITEM	DESCRIPTION	COST
1	**FLASH**	**32GB flash disk**	**Tshs 20000/=**
2	**MODERM**	**For connecting to internet**	**Tshs 80,000/=**
3	**INTERNET BUNDLE**	**For three months Tshs20000/=@month**	**Tshs 60000/=**
4	**STATIONERY**	**1: Questionnaire papers printing About 25 papers Tshs150/=@page. 2: proposal printing Tshs150/=@page**	**Tshs 10000/=**
5	**MOBILE CREDIT**	**For three months Tshs 10,000/=@month**	**Tshs 30,000/=**
6	**DOMAIN**	**For three-month Tshs 25,000/=**@month	**Tshs 75,000/=**
	TOTAL		**Tshs 275000/=**

Table 5.1 Materials and Cost